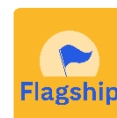


18 Sep 2026 18:03:07 ET | 35 pages

US Rates Weekly

Only hawks outside the window



CITI'S TAKE

Chair Warsh was unquestionably hawkish in our view, but such action should limit downside tails for the long-end of the curve for now. Core long positions face two challenges ahead: 1) Avoiding a rate hike in October and 2) weathering the seasonally strong Jan and Feb core PCE prints. We are concerned on both fronts and close our long 20y bond position at a loss. For investors who can withstand this volatility, we do still see value especially into the November refunding. We update our rates forecasts and see 10y USTs at 5% by year-end even if the FOMC keeps rates on pause at the October meeting.

USTs — The latest flow of funds data reports hedge fund basis books have shrunk the most since COVID. We outline why this is not as large of a risk as the popular narrative suggests.

Short-end — SOFR has continued to fix below IORB this past week. We show how reserves have been a larger driver of repo rates compared to T-bill outstanding recently and we think this suggests we're in a lightly abundant reserves environment. We present a framework to estimate SOFR/FF for the rest of the year, and we find that current market pricing looks bearish compared to our estimates. Lastly, we discuss what Fed Vice Chair for Supervision Bowman's timeline for GSIB reform means for year-end liquidity conditions.

Inflation — We examine the downside risk for inflation expectations after the hawkish FOMC. We continue to like our short 2y breakevens position with the energy hedge as we see a lot of room for ex-gasoline inflation expectations to move lower from here.

Vol — Rates vol has been and should continue to be driven by the expected pace of the Fed's hiking cycle. The 2s5s rate curve could flatten meaningfully in an overly hawkish/policy error scenario, and we recommend buying SFRH7 put against 2y midcurve SFRH9 put as a cheap front-end bear flattener tail hedge.

Please join us for a 15-minute call on Monday September, 21, at 10am NY time/3pm London time when we will summarize key views for the week ahead: [Global Rates & FX Weekly Call](#)

Note: Clients subject to MIFID II unbundling should have a full-service research agreement in place to participate. If unsure about your agreement, please check with your in-house, sell-side research coordinator, or with the event organizer

Note: Futures trading involves substantial risk of loss.

Jason Williams^{AC}

+1-212-723-1837

jason1.williams@citi.com

Mike Chang^{AC}

+1-212-816-8718

mike.chang@citi.com

Raghav Datla^{AC}

+1-212-723-4813

raghavendra.varma.datla@citi.com

Alejandra Vazquez Plata^{AC}

+1-212-723-0486

alejandra.vazquezplata@citi.com

*With thanks to
Simran Bedi*

See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

Contents

Contents	2
Summary of Rate Expectations	3
Overview: Only hawks outside the window	4
Can the FOMC out hawk the forwards?	4
Taking off our long bias in 20s	6
Updating our forecasts: 10y at 5% by YE and lower in 2027	6
USTs: The basis apocalypse has been greatly exaggerated	8
Keep calm, basis books are returning to the grave	8
Signals that the basis trade is dying out	9
Why have dislocations been eaten up?	9
Limited risks to market function from a basis resurrection	10
Short-end: SOFR more likely to stay below IORB for now	12
Interrogating inflation: Stay short	15
Another reason for selling inflation	15
Vol: Cheap front-end bear flattener	18
US Rates Strategy Model Portfolio Update	21
Outstanding Trade Recommendations	22
Appendix I: Model Portfolio Closed Trades	24
Appendix II: Treasury Relative Value	25
Appendix III: Swap Fly Relative Value	26
Appendix IV: Front End Monitor	27
Appendix V: FX Hedged Yield Pickup	28
Appendix A-1	29

Summary of Rate Expectations

Figure 1. Summary of views

Factor	View	Recommended Positions
Duration	Neutral	SFRZ6 1x2 put spread, SFRZ6/Z7 flattener vs long FFX6
Yield Curve	Flatter	SFRH7/H9 conditional bear flatteners, Buy 1y CMS 30s/5s curve cap contingent SPX, 1y fwd 10s30s bear steepener via payers
Swap Spreads	Wider	Long 5y swap spreads, 10s20s spread steepener
Gamma	Short	Long costless 3m2y receiver ladder
Vega	Neutral	
Inflation	Tighter	Short 2y BE hedged with gasoline
Front-end	Tighter	Pay 1y1y SOFR/TONAR

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Note: See Figure 39 for trades pricing. The Buy 1y CMS 30s/5s curve cap contingent on SPX trade idea is owned by **Mike Chang**.

Note: Futures trading involves substantial risk of loss.

Figure 2. Rates forecasts for 2026

US Rates	Spot	Forwards	Base Case	Bull Case	Bear Case
2y	4.66%	4.78%	4.80%	4.30%	5.30%
5y	4.78%	4.84%	4.85%	4.45%	5.30%
10y	4.93%	4.99%	5.00%	4.65%	5.30%
30y	5.28%	5.30%	5.30%	5.05%	5.55%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research.

Note: Spot and YE forwards as of September 17, 2026.

Figure 3. Supply forecasts for 2026

2026	FRN	Nominal Treasuries							TIPS			Gross Supply to pvt (A)	Redemption (from pvt) (B)			Net Issuance (A-B)	Treasury Buybacks (C)		Net Issuance incl. buybacks (A-B-C)
	2y	2yr	3yr	5yr	7yr	10yr	20y	30yr	5y	10y	30y		Nominals	TIPS	FRNs		Liquidity Support	Cash Management	
Jan-26	30	69	58	70	44	39	13	22	0	21	0	366	40	56	0	270	10	0	261
Feb-26	28	69	58	70	44	42	16	25	0	0	9	361	232	0	84	45	12	0	33
Mar-26	28	69	58	70	44	39	13	22	0	19	0	362	313	0	0	49	7	30	12
Apr-26	30	69	58	70	44	39	13	22	26	0	0	371	184	31	86	70	6	45	19
May-26	28	69	58	70	44	42	16	25	0	19	0	371	83	0	0	288	9	0	279
Jun-26	28	69	58	70	44	39	13	22	24	0	0	367	314	0	0	53	7	25	21
Jul-26	30	69	58	70	44	39	13	22	0	21	0	366	173	39	86	68	11	0	58
Aug-26	28	69	58	70	44	42	16	25	0	0	8	360	229	0	0	131	16	0	115
Sep-26	28	69	58	70	44	39	13	22	0	19	0	362	187	0	0	175	17	25	133
Oct-26	30	69	58	70	44	39	13	22	26	0	0	371	42	44	0	285	29	0	256
Nov-26	28	69	58	70	44	42	16	25	0	19	0	371	378	0	86	-93	17	0	-110
Dec-26	28	69	58	70	44	39	13	22	24	0	0	367	198	0	0	169	16	25	128
Total	344	828	696	840	528	480	168	276	100	118	17	4,395	2,373	169	342	1,511	158	149	1,204

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Note: Liquidity support buybacks utilize securities of all maturities across nominals and TIPS, while cash management buybacks generally utilize one-month to two-year maturities. We assume all Fed reserve management purchases are T-bills. If the issue date of coupons falls on weekend of month end, the supply doesn't roll over to next month in this table for illustration purpose.

Overview: Only hawks outside the window

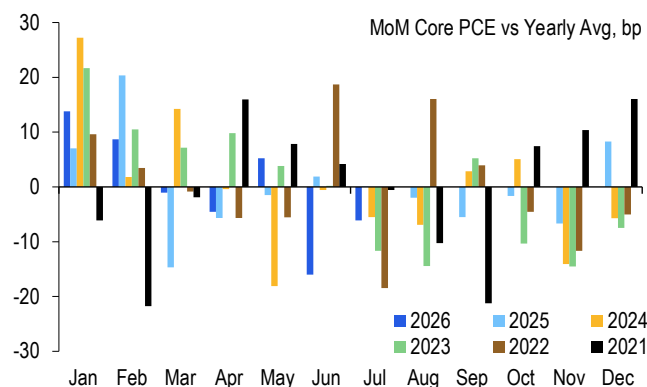
Jason Williams

Can the FOMC out hawk the forwards?

This week's FOMC was decidedly hawkish, in our view. The reaction function appears to be squarely year-over-year core PCE which Chair Warsh previously emphasized at Jackson Hole (see [US Rates Strategy - Remain long 20s after Chair Warsh's speech](#)), although he did mention 6mo core PCE at times during the press conference. His response on why they hiked this week, as opposed to July, further reinforced his hawkish view. He saw that "inflation summer trends weren't passing the test" and that there has been "very little information since" that would change that judgment. That is a powerful response when core goods, OER, and rent inflation looked very benign at the last CPI print. More striking was his emphasis on headline inflation having an impact, stating that the "geopolitical situation has changed," effectively emphasizing higher oil prices, and he had a broad view that the "economy has strengthened" as well. Finally, he did not materially mention his task forces which, at the June meeting, appeared to be a reason to delay a hiking cycle. That delay is decidedly over.

This is unlikely to be the Fed's only hike if Chair Warsh is squarely focused on YoY core PCE (note that our economists are calling for the Fed to remain on pause until easing policy rates in H2 2027; see [US Economics - FOMC - Hawkish 25bp rate hike](#)). The challenge for markets, and the Fed, will be that January and February core PCE prints tend to have strong residual seasonality (Figure 4). The Fed may still want to hike into December and March due to uncertainty around this seasonality, even if Q4 inflation prints look more benign.

Figure 4. Typically, there is residual seasonality in the January and February core PCE prints



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Figure 5. Rates are moving very restrictive as front-end inflation swaps get pushed lower



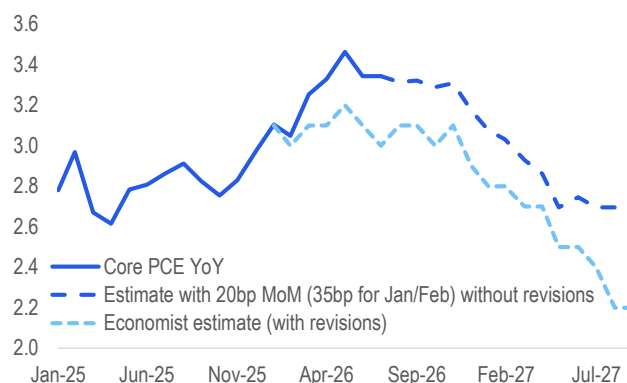
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

Forwards are pricing too much Fed restriction, in our view, either relative to where the Fed is going or where they *should* go. What we mean is 1y1y RYs are now at post-Covid highs even though inflation is clearly falling on most metrics (Figure 5). And if the FOMC decides to hike at a quarterly cadence, perhaps the modal outcome, then it will be hard to out hawk the forwards due to inflation base effects. In Figure 6, we report a simple projection of core PCE YoY into 2027. We do this to highlight that base effects will help lower the YoY measure but not until the middle to late 2027. Our estimate is an illustrative example of a series of 20bp core MoM prints and we assume January and February will print 35bp each. Note that we do

not report a bespoke estimate for the upcoming core PCE revisions. Instead, we provide our economist forecasts for YoY which incorporate the changes in light blue and include their MoM forecasts (our economists estimate the revision at ~20bp as described in [US Economics - Inflation Weekly – August inflation coming down to the wire\(less\)](#)).

We continue to be received the October FOMC meeting, as we expect the Fed to hike at a quarterly cadence until core PCE gives them an all clear at some point in 2027. Indeed, the SEP dot plot was not overly hawkish, with only four members wanting to hike at the remaining two meetings. This implies that it will be challenging to gain a broad consensus for hikes in October. To be clear we do not know what Chair Warsh is thinking. Still, the rate cuts in 2024 and 2025 were well telegraphed by the September dot plot (Figure 7). Presumably the Fed will not want multiple dissents with the US election the following week. Still, a 0.3% MoM core CPI, released on October 14, could bring about a more unified committee for a hike. Presumably, a 0.1% MoM core CPI print would alleviate near-term concerns. Finally, the FOMC will see the core PCE revisions, released September 30, which should be a dovish catalyst. The SEP core PCE forecasts did not indicate these revisions, in our view, so this should support an October skip.

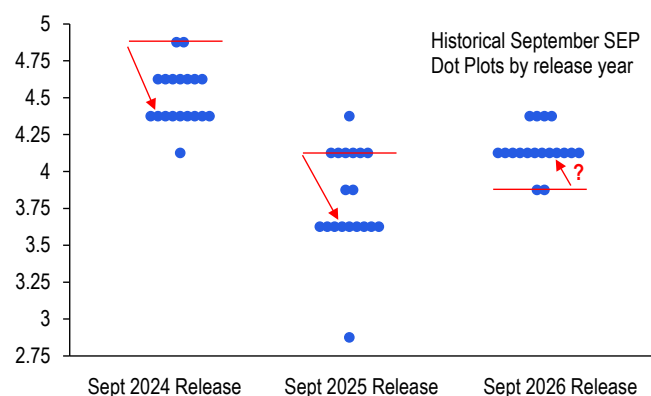
Figure 6. YoY Core PCE will fall dramatically in mid-to-late 2027 due to base effects and slowing of core components. But prints are likely to remain strong through Q1 putting hawkish pressure on the FOMC.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Figure 7. The three rate cuts in 2024 and 2025 were well telegraphed by the dot plot. The September 2026 SEP is not that supportive of an October and December hike.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

Below, Mike discusses how conditional bear steepeners on SOFR futures look very attractive here for investors looking to protect themselves from an October hike, which presumably would open the door for multiple hikes in a row. Buying a put on SFRH7 but funding that by selling a midcurve puts limits terminal downside tails in a rally. Alternatively, in Raghav's inflation section, he discusses how there is still room for front-end market expectations of core inflation to move lower still. For instance, 1y1y inflation swaps are down almost 10bp from the highs of the week and we see fair value below current levels. Both structures position for a more hawkish Fed with limited downside tails. *Note: Futures trading involves substantial risk of loss.*

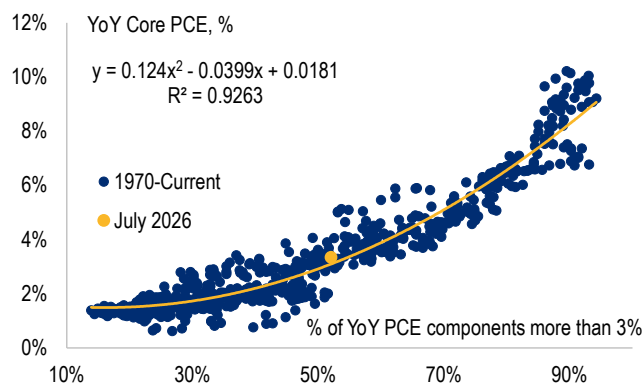
The risks for an October hike are serious, especially if the FOMC looks through the PCE revisions. For instance, the revisions will not improve Chair Warsh's disaggregated PCE metric which he mentioned again Wednesday, stating "Too many categories are still posting increases love 3% on both the 6 and 12-month

basis" (Figure 8; we did a deep dive on this measure in [US Rates Weekly - Is a hike the antidote?](#)). To be clear, core PCE and Warsh's component measure are exceptionally correlated. Note that this measure, due to the randomness of the components, many of which have very tiny weights, has a noise floor, historically speaking (Figure 8), and October remains at risk from higher oil prices, especially if the Middle East conflict expands heading into the US election.

Taking off our long bias in 20s

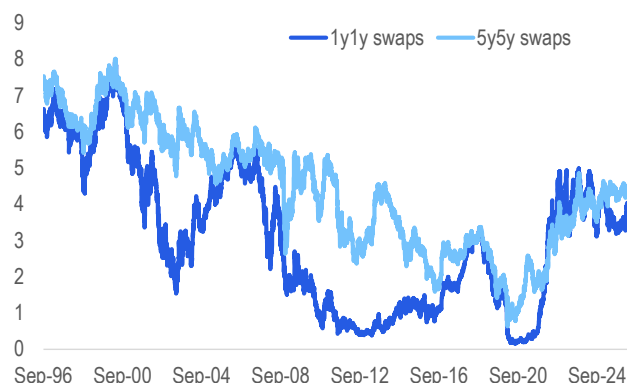
We now close our long 20y bond position at a loss of \$570k at a price of 95-09, as we are close to the stop-out level although we did not hit it yet (pricing as of 2pm on 9/18/26, see [US Rates Strategy - Getting long 20y after today's buyback announcement](#)). Still, we continue to see value in the point and still favor longs in the back end compared to the front end (e.g., flatter curve). We struggle to see a near-term bullish catalyst following this FOMC even though we find valuations exceptionally attractive. While much ink has been spilled on deficits and the worsening fiscal picture for bonds, at the end of the day, the UST selloff has been driven by 1y1y moving higher. And we struggle to see that story coming to a close in the very short term.

Figure 8. Disaggregated PCE generally follows core PCE. The revisions will lower core PCE but not help the disaggregated metric.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Figure 9. 1y1y swaps have now surpassed 5y5y. Could this be a sign that 5y5y has reached the local highs?



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

For those who can withstand this volatility, and take a longer-term view, we still see value. As we explained in our last weekly, 5y5y does not have to move higher in a hiking cycle, which is the case so far post FOMC ([US Rates Weekly - Is a hike the antidote?](#)). Further, the fact that the 1y1y/5y5y swaps curve is now inverted likely means we are closer to the end of the neutral debate than the beginning (Figure 10). Or, in other words, the market does see this FOMC as restrictive, and that implies they may not need to beat front-end forwards by much.

If the FOMC holds rates steady in October, the market may quickly price in a quarterly FOMC cadence which limits how many rate hikes are achievable before Q3/Q4 2027 (Figure 10). The November refunding, which is the week after the October FOMC, also gives Treasury Secretary Bessent a chance to shift issuance away from the long-end, which would be supportive of 20y bonds.

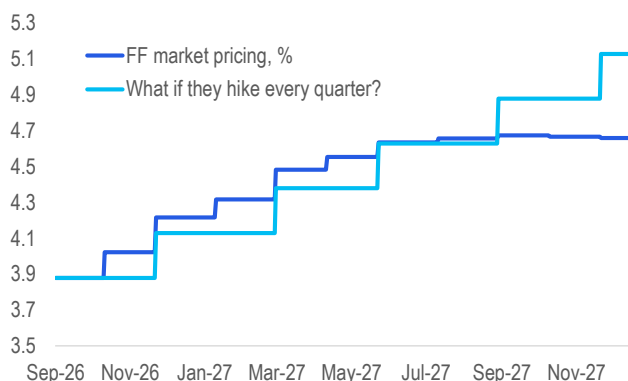
Updating our forecasts: 10y at 5% by YE and lower in 2027

We now modify our yield forecasts following the September rate hike. Typically, we are slow moving with our forecasts to limit chasing price action, but the world is

now fundamentally different today compared to the July FOMC. We assume Fed rate hike pricing will remain until YoY core PCE starts to normalize next year. For the rest of 2026, we expect the price of oil and the path of YoY core PCE will drive the rates market, neither of which paint a good picture for yields in the short-run.

Our 10y YE26 forecast is now 5.0% (Figure 11). Mechanically, we assume the FOMC will pause rates at the October meeting and hike rates at the December meeting. We expect the market to price in further tightening with 1y swaps realizing towards 4.75% by YE26. We also assume oil forwards will roughly be realized. Our bear scenario, where 10y USTs reaches 5.3%, assumes the FOMC goes in October and December and 1y swaps would price in further tightening to 5.25% by YE26. There is a chance that 2s5s would be inverted and our earlier mentioned 1y1y/5y5y would likely start pricing in a policy mistake in this scenario (hence skew risk is for a flatter curve). Our bullish scenario, where 10y USTs reaches 4.65%, assumes the Fed keeps rates on pause but we enter 2027 pricing in rate hikes with 1y swaps at 4.2% by year-end. One possible reason would be a large drop in the price of oil combined with Chair Warsh focusing more on 6mo annualized core PCE, which should look good by December, before moving higher in January and February.

Figure 10. There is a limit to how high the Fed can hike, relative to forwards, if they go at a quarterly cadence



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Figure 11. See below for our new forecasts

YE 2026 forecasts

UST	Base	Bull	Bear
2y	4.80	4.30	5.30
5y	4.85	4.45	5.30
10y	5.00	4.65	5.30
30y	5.30	5.05	5.55

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

We also clean up our portfolio, which has been heavily bullish duration. Thankfully, we had some paid positions as a hedge, via a paid SOFR U6, which offsets some of the loss. That trade netted us +\$737k at 96, up 29.5 ticks (pricing as of 2pm as of 9/18/26, see [US Rates Weekly - Hedging our bullish lean into CPI](#)). We close our core steepener position, which was down 66bp (1y 5s30s swap steepener pricing as of 2pm as of 9/18/26, see [US Rates Weekly - Trade picking 2025](#)). We expect further flattening tail risks in the short term. *Note: Futures trading involves substantial risk of loss.*

USTs: The basis apocalypse has been greatly exaggerated

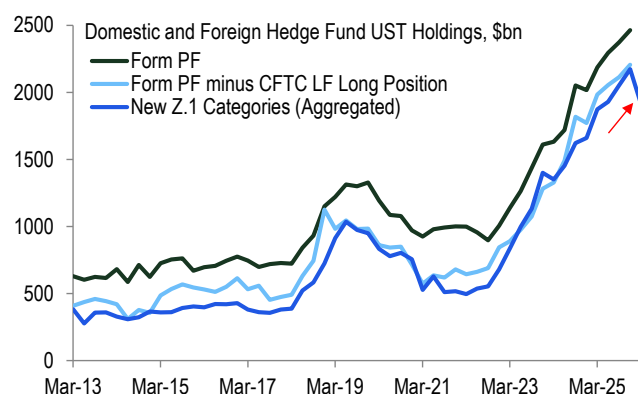
Keep calm, basis books are returning to the grave

Last Friday, the Fed released Z.1 data showing hedge fund Treasury cash positions declining the most since COVID (Figure 12). This is likely a decline in basis books. The latest Z.1 report was the first with newly added categories “Domestic Hedge Funds” and “Foreign Hedge Funds” in the main financial accounts. Domestic and foreign hedge fund holdings of USTs fell a combined ~\$260bn from Q4 2025 to \$1,914bn in Q1 2026. Foreign hedge funds are a majority of the combined holdings with \$1,806bn foreign and \$108bn domestic. Note that Z.1 reports market value.

As a reminder, hedge funds typically enter long basis trades (sell futures and buy off-the-runs) for the carry provided by richer futures compared to a corresponding cheapest-to-deliver (CTD) or cheap off-the-run paper. So, we can gauge the size of these books on either hedge fund long cash positions or short futures positions. Previously, there was lagged, aggregated hedge fund holdings data through SEC’s Form PF and various Z.1 tables (that used underlying Form PF data). Hedge fund holdings were in the main Z.1 financial accounts data integrated into bulk categories, but the hedge fund data itself was e. There was also domestic only hedge fund data in the supplementary table of Z.1, and post-GFC the Fed added the “Enhanced Accounts,” which included an aggregated (foreign + domestic) hedge fund balance.

Unlike previously, Z.1 now reports UST holdings excluding derivative longs (i.e. futures). In Figure 12, we show the new Z.1 data, updated to Q1 2026, against Form PF data (which is up to Q4 2025). We can reconcile the new Z.1 data by roughly subtracting CFTC levered fund longs (a proxy for hedge fund future positioning) from Form PF. Unfortunately, even given this new data, the size of basis books are still challenging to estimate, as the long cash position of hedge funds is not entirely basis trades.

Figure 12. New Z.1 data shows a decrease in hedge fund cash UST holdings.



© 2026 Citigroup Inc. No redistribution without Citigroup’s written permission.
Source: Citi Research, SEC, Federal Reserve

Figure 13. CFTC data shows levered fund gross shorts are decreasing, indicating shrinking basis books.



© 2026 Citigroup Inc. No redistribution without Citigroup’s written permission.
Source: Citi Research, CFTC

In June, the Fed published a [paper](#) trying to identify the size of basis books. They argue that other types of trades (like swap spreads, steepeners/flatteners, cash, etc.) would be a significant part of the holdings, too. They estimated that as of September 2025, basis trade volume is around \$830bn and swap spread volume is

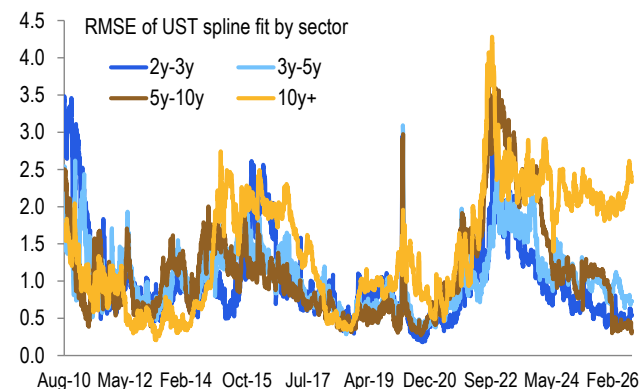
about \$305bn. It has always been imprecise to “back out” the size of basis trades, as trade types are not mutually exclusive and hard to define. For example, a 1y paper “basis” trade could be against swaps, not futures. Nevertheless, basis book carry has been falling and funds likely added some short-dated swap spread longs to improve carry. Recently, we have seen a decrease in levered fund gross shorts per the CFTC dataset, and it appears that basis trade activity has been shrinking (Figure 13). Relative value opportunities in the Treasury market have become sparse but this may not necessarily be worrisome.

Signals that the basis trade is dying out

Basis books are declining as dislocations and volatility have fallen materially over the past few years. The consequences are visible in the below.

1. **Dislocations in off-the-runs are declining.** In recent months, we have seen a decline in the RMSE of our UST spline fit in various sectors (Figure 14). This means the variance of the rich/cheap of off-the-runs has been lower, suggesting less opportunities to find cheap paper and lower profits in the cash-futures basis.
2. **Treasury buyback operations have less “cheap” sellers.** Last week, the Treasury conducted its first buyback operation after announcing increased sizes in the 10y+ sectors. The maximum operation size was raised to \$6bn but only \$5.187bn was utilized. Given that buybacks are controlled by the sellers, who offer a certain price that Treasury deems “cheap enough” to buy, we question the amount of cheap off-the-run collateral in the market
3. **Repo volumes are falling.** Repo this year has been quite calm and there have not been signs of large sales. As we have written previously, repo markets have not shown any pressure in recent weeks, even with increases in T-bill outstanding (see [US Rates Weekly – Can Secretary Bessent and Chair Warsh save the curve?](#)).

Figure 14. Dislocations in off-the-runs have been declining recently.

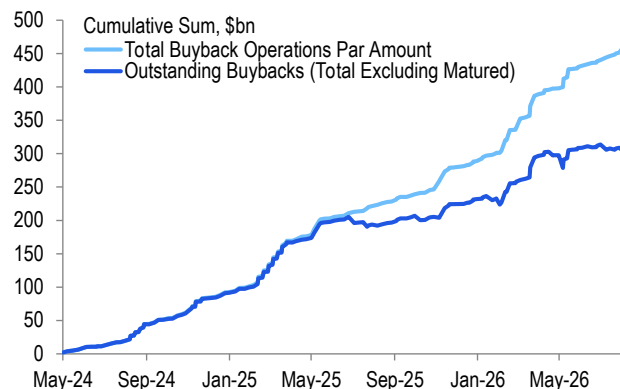


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Note: We exclude deep oftr 30y bonds in our calculations.

Source: Citi Research

Figure 15. Buybacks have taken a significant amount of off-the-runs out of the market.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Treasury

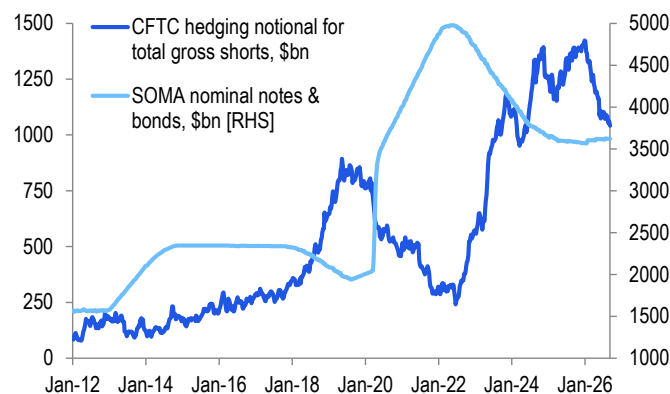
Why have dislocations been eaten up?

We do not think the decline in the basis trade is worrisome for the Treasury market, despite some popular narratives. The lack of growing basis books indicates that

there are not many wholesale UST sales of off-the-runs. More specifically, we see the following reasons for shrinking basis books:

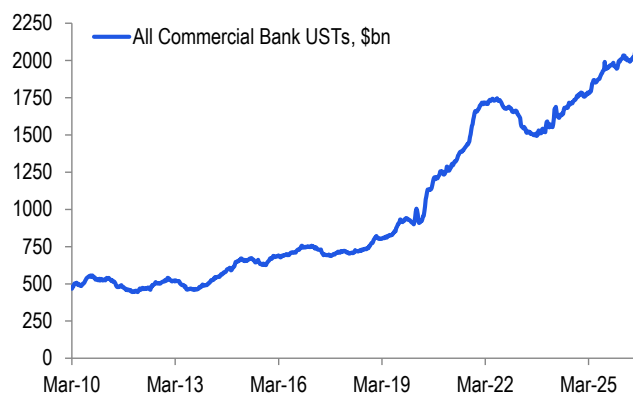
1. **Asset manager demand for futures is falling.** When basis books were expanding, asset managers had been moving overweight duration via leverage, which can richen futures. Presumably, the increased uncertainty around inflation and the Fed has kept asset managers at or underweight their duration benchmarks.
2. **Buybacks have removed a sizable amount of old CUSIPs from the market.** Since the start of the buyback program, Treasury has bought a total par amount of \$480bn USTs through its operations, as seen in the light blue line in Figure 15. The dark blue line in this figure would be the amount of additional off-the-runs in the market if not for buybacks. As a reminder, the buyback program pulls forward redemptions, so when Treasury buys a security back, it is instantly “deleted”. The dark blue line is the total cumulative buyback operation amount accounting for the original maturity of those securities. Given this, there would be ~\$300bn more off-the-runs in the market today, if not for buybacks.
3. **The Fed stopped QT last year.** The basis trade grew during both previous QT episodes during 2017-2019 and 2022-2025 (Figure 16). The Fed stopped QT last year, as repo pressures grew and is unlikely to restart in the short term.
4. **Bank demand for USTs has been increasing steadily.** We have seen an acceleration in bank demand of USTs (Figure 17). Banks could be more comfortable converting their cash into USTs after QT ended. We also think banks have been leaning into UST ASWs for the attractive carry and rolldown after the implicit Fed backstop in 2025 via RMPs.

Figure 16. Basis trades seem to pick up during QT, which we do not see restarting in the near term.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, CFTC, Bloomberg

Figure 17. Increased bank demand for USTs is also taking paper out of the market.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

Limited risks to market function from a basis resurrection

Frequently, the cash-futures basis trade is mentioned as a key risk to the Treasury market. First, it is important to understand that the basis trade supports the off-the-run Treasury market, it does not help the Treasury market warehouse duration. Or, in other words, if the basis trade was stopped out, it would have some light impact on market function, but it would not mean yields would need to move higher. Especially since these off-the-runs are hedged with either futures and/or

swaps. 2020 is often used as an example for its effects on the market during basis unwinds, however this is a poor analog for true market effects due to the sheer amount of UST sales by other participants, which the Fed had to step in to buy (Figure 16). These sales are what truly broke the Treasury market in 2020, not entirely basis trades.

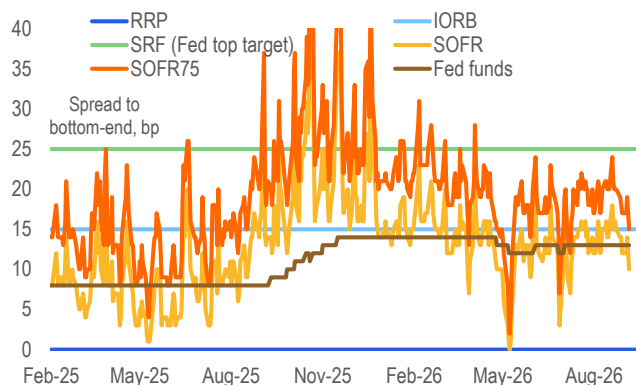
Even so, the large move in yields this year has not resulted in any material sales of Treasuries, as evidenced above. Furthermore, the market is positioned to absorb risks of selling. The buyback program, which we have recently seen can be adaptive, limits the amount off-the-runs in the market, and as we saw last week, still has some leeway. Also, the ample repo markets right now could easily digest large sales in USTs. Rather than signaling risk, the shrinking opportunity set suggests that underlying Treasury demand may be stronger than we think. Or, in other words, the narrative would likely be more concerning if dislocations in off-the-runs were growing and the size of RV books was increasing, at least as far as market functioning.

Short-end: SOFR more likely to stay below IORB for now

Alejandra Vazquez Plata
Jason Williams

SOFR fixes have continued to soften after the August month-end spike and for the past two weeks SOFR has been fixing at or below IORB (see Figure 18). As we discussed last week, we think that reserves are once again in a lightly abundant regime and therefore SOFR fixes have been more correlated with changes in reserves than with changes in T-bill outstanding (see [US Rates Weekly - Is a hike the antidote?](#)). In Figure 19, we regress the daily level of SOFR/IOB against private T-bill outstanding minus reserves using data for the first half of the year and use those coefficients to estimate SOFR/IOB since July. We find that SOFR/IOB has realized lower compared to what we would have expected based on the level of private T-bills minus reserves. In other words, the increase in private T-bill outstanding in July and August didn't push SOFR/IOB higher. Also, in Figure 19, we repeat this exercise but instead use only the level of reserves as independent variable and find a generally better fit.

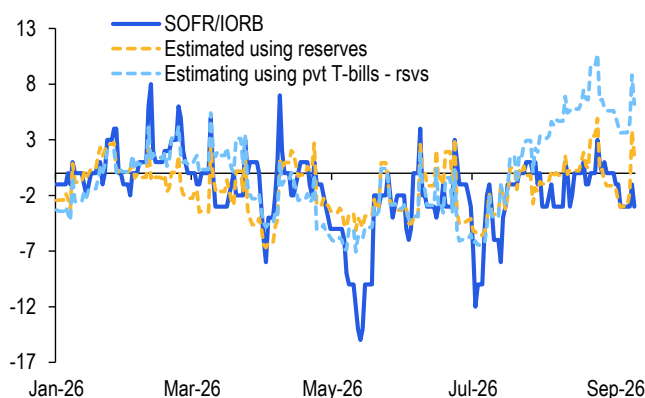
Figure 18. For the past two weeks, SOFR has been fixing at or below IORB, we think this has been driven by the relative stability of reserves.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

Figure 19. The recent changes in private T-bill outstanding have not resulted in similar changes in SOFR/IOB.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

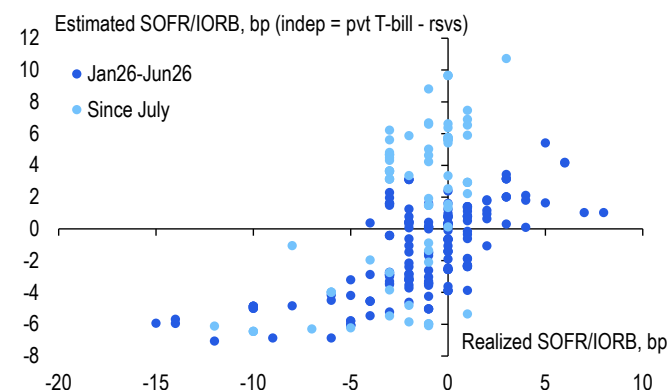
Note: we use daily data from 1/8/26 to 6/30/26 to fit a levels regression and control for coupon settlement dates.

Source: Citi Research, Bloomberg

To better visualize this disparity, we plot SOFR/IOB fixings against our estimated SOFR/IOB level using private T-bills minus reserves and using only reserves as independent variables. We show the contrast between the regression data (first half of the year in dark blue dots) and the estimated sample (since July 2026). While the light blue dots in Figure 20 are elevated, the yellow dots in Figure 21 are more closely aligned with the 45-degree line.

We think that reserves will continue to be the main driver of repo rates for the rest of the year, or in other words we expect this lightly abundant reserves regime to be the norm for the upcoming months. With RMPs likely on pause for the rest of the year, the TGA balance will be the main determinant of the level of reserves going forward. We think the TGA will generally be between \$850bn and \$950bn for the rest of the year, however we'll likely see a temporary spike towards \$1tn in the last few days of October, and we expect to see a decline in TGA during December as Treasury guided towards an \$850bn target by year-end.

Figure 20. Our preferred liquidity needs proxy (private T-bill outstanding minus reserves) suggests that SOFR/IORB fixes since July should have been more elevated.

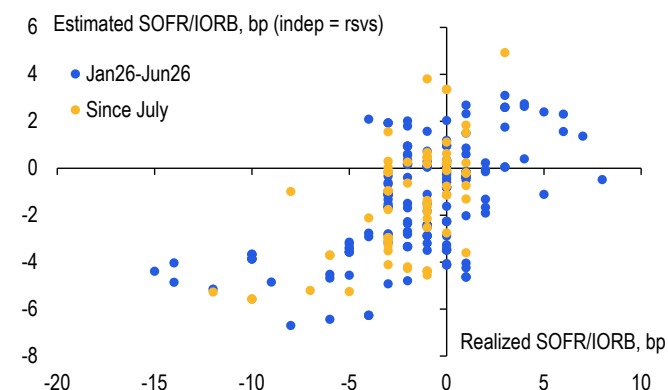


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Note: we use daily data from 1/8/26 to 6/30/26 to fit a levels regression and control for coupon settlement dates.

Source: Citi Research, Bloomberg

Figure 21. In contrast, it seems that reserves have been a more relevant driver of SOFR/IORB fixes over the past few weeks.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Note: we use daily data from 1/8/26 to 6/30/26 to fit a levels regression and control for coupon settlement dates.

Source: Citi Research, Bloomberg

How to think about SERFF

We use the two frameworks discussed above to estimate SOFR/FF contracts for the rest of the year. We start with the regression coefficients of SOFR/IORB of the regressions described above and use our weekly forecasts for changes in private T-bill outstanding and reserves levels to estimate a baseline weekly change in SOFR/FF. We then account for coupon settlement dates and weekend rollovers and put it all together to calculate a daily and monthly average SOFR/FF level. We show our results in Figure 22. Our estimates show that the market's expectations of SOFR/FF are more pessimistic compared to what our models imply. For example, our models suggest that SOFR will generally fix below FF in both October and November. Since our framework is susceptible to the starting level of SOFR/FF used, the recent soft SOFR prints are likely skewing our results towards a more positive SOFR/FF level. Therefore, we think of these estimates as 'best case' scenarios for the rest of the year. Uncertainty around the Fed's near-term reaction function could also keep SOFR relatively low as more cash is allocated to repo.

Note: Futures trading involves substantial risk of loss.

Lastly, we estimate that the current level of Dec SOFR/FF (-3.5bp) implies an average level of \$2.78tn reserves through the month, which we largely disagree with, and instead we think reserves in December are more likely to range between \$2.9tn and \$3tn. We think that reserves in December will be supported by a move lower in TGA towards \$850bn, and by relatively calm liquidity conditions at year end. As we discuss below, we think banks face different regulatory incentives this year compared to last year, and this can result in better liquidity availability than what markets are currently expecting. On the flip side, a tail risk that could push our SERFF estimates closer to current market pricing for December is a significant increase in T-bill issuance if Treasury decides to shift long-end coupon issuance into the front-end. We think we could get more clarity on this when the dealers' refunding questionnaire is released on October 16.

Note: Futures trading involves substantial risk of loss.

Figure 22. The SOFR/FF strip seems to be pricing in a lower level of reserves and/or more private T-bill supply in December.

Estimated SOFR/FF, bp

Month	Pvt T-bills - rsvs	Reserves	Futures
Sep-26	-0.05	-0.05	-0.25
Oct-26	1.35	1.65	-1.25
Nov-26	0.50	1.95	-2.00
Dec-26	-0.10	2.00	-3.50

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Note: Futures trading involves substantial risk of loss.
We use daily data from 1/8/26 to 9/18/26 to fit a SOFR/IOB levels regression and control for coupon settlement dates.

Source: Citi Research, Bloomberg

Figure 23. We generally find that SOFR/IOB normalizes after a change in the FF range.

SOFR/IOB after FF change

	Sep-24	Nov-24	Dec-24	Sep-25	Oct-25	Dec-25	Sep-26
-5	-7	-4	-3	1	9	2	-3
-4	-7	-8	-5	2	9	3	-3
-3	-2	-8	0	11	12	5	-3
-2	-2	-9	-3	-1	16	3	-1
-1 (FOMC day)	-7	-8	-8	-2	12	0	-3
0 (day after)	-8	-5	-10	-1	14	1	-5
1	-7	-5	-10	-1	32	2	
2	-7	-6	-9	-1	23	10	
3	-6	-7	0	-3	10	4	
4	-6	-8	13	-2	1	4	
5	-7	-8	6	3	2	1	

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

A note on FF pass through

We have received a few questions on the pass-through effect on SOFR from this week's FOMC hike. In Figure 23, we show the last 2 years of FOMC dates when there was a policy change and the level of SOBR/IOB around those dates. The table shows that SOFR/IOB generally stayed within a similar range before and after the FF change. Further, since some of these meetings took place close to a quarter-end or amidst the repo stress from last year, it's hard to argue that the change in FF lead to a meaningful change in SOFR/IOB.

What's next in the regulatory agenda

This morning in prepared [remarks](#), Fed Vice Chair for Supervision Bowman announced she expects regulators will finalize the "improvements to the global systemically important bank (GSIB) surcharge" before year-end. We discussed our takeaways from the proposal in [Short-End Notes - Repo to remain calm into quarter-end](#). Usually, regulators allow banks to adopt a new rule as soon as it is finalized. So, we expect banks to treat this year-end with the easier GSIB calculation parameters, as even a 2027 adoption of the rule would come with a 2-year lookback period. As a reminder, a bank's surcharge for a given CY is the minimum of the surcharge calculated in the CY immediately prior or the surcharge calculated in the prior two CY.

Once the GSIB reform is finalized, we think regulators will focus on reforming the Liquidity Coverage Ratio (LCR) calculation. As we have discussed before, we think that LCR reform can help the Fed reduce banks' demand for reserves, which in turn would enable the Fed to reduce the size of their balance sheet (see [FX Basis Quarterly - Could next week's FOMC jeopardize the basis?](#)).

Raghav Datla

Interrogating inflation: Stay short

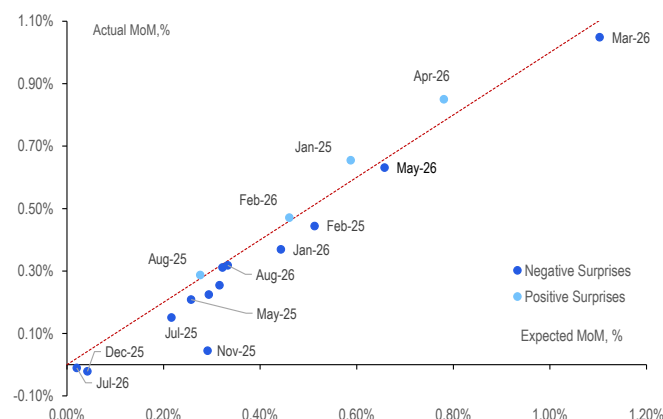
We examine the downside risk for inflation expectations after the hawkish FOMC. We continue to like our short 2y breakevens position with the energy hedge as we see a lot of room for ex-gasoline inflation expectations to move lower from here.

Another reason for selling inflation

A hawkish Fed joins a growing list of reasons to be short CPI inflation in the US.

The hawkish FOMC meeting this week is the latest (and likely the most decisive) among a growing list of drivers that will push US inflation expectations lower in the near term. While the August CPI print was higher than expected, the headline inflation print was still less than what the inflation fixing markets were pricing (Figure 24), and it was completely driven by a one-time increase in wireless services component (see [US Rates Weekly - Is a hike the antidote?](#)). Even with that surprise, YoY core inflation decreased slightly in August to 2.45% and ex-gasoline inflation has declined to 2.22%. However, markets are currently pricing for an increase in ex-gasoline inflation to 2.8% over the next year (Figure 25), and we continue to fade that via our short 2y breakevens position (hedged with gasoline futures) which we initiated in July (see [US Rates Strategy - Shorting 2y breakevens](#)). That trade has moved in our favor since then, but there is still lot of room left. *Note: Futures trading involves substantial risk of loss.*

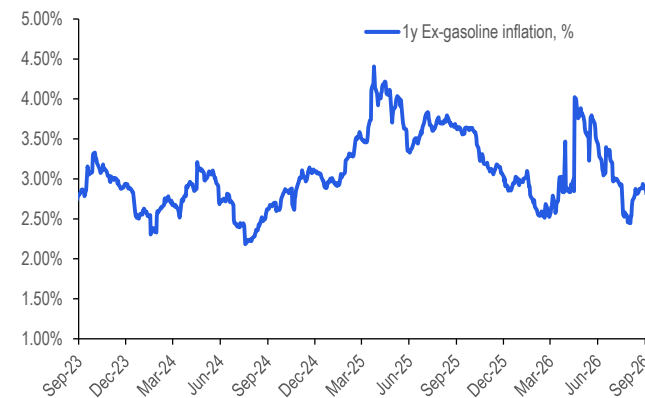
Figure 24. 15 of the last 19 headline CPI prints were less than what inflation markets were pricing heading into the release.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

Figure 25. Inflation markets are still pricing for close to 2.8% ex-gasoline inflation over the next year.

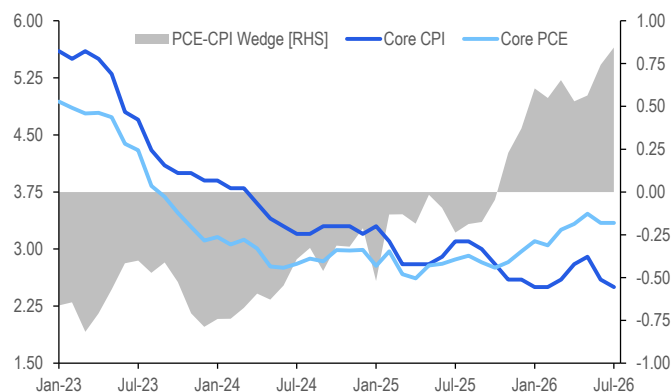


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

Reaching target level core PCE will likely require below target core CPI. The fact that the Fed is starting to fight inflation when core CPI inflation has already moved close to the target levels only increases the downside risk for CPI inflation going forward. Chair Warsh's Jackson Hole speech and the press conference this week made it clear that the Fed are clearly targeting YoY core PCE. That increases the downside risk for CPI inflation because, the more rate sensitive parts of the inflation basket like shelter have a bigger weight in CPI than PCE. This has already led to a widening of the wedge between YoY core CPI and core PCE in recent months (Figure 26). The components contributing to the strength in core PCE like Portfolio management fees, Computer software and accessories etc are not that sensitive to rates. So, bringing YoY core PCE to 2% could mean that inflation in rates sensitive components like shelter will have to be brought down significantly below 2%. That could lead to a sharp decline in YoY core CPI in coming months.

Figure 26. The wedge between core PCE and CPI inflation has been widening this year.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Figure 27. Percentage of metro areas with greater than 4% OER inflation has decreased sharply in recent months.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, BLS, Haver Analytics

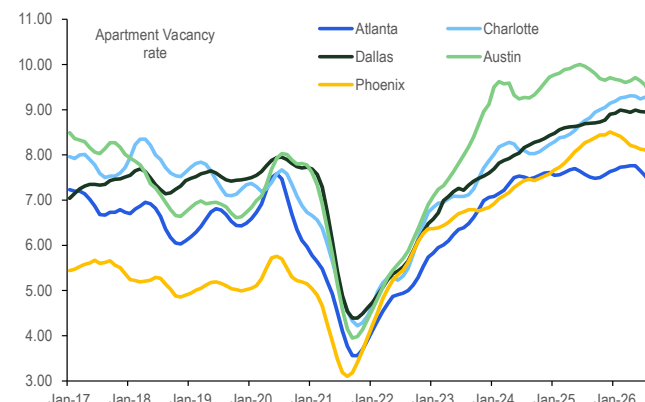
Shelter inflation will continue to move lower and will be a drag on ex-gasoline CPI inflation. The main reason behind ex-gasoline inflation staying above 2% over the last year is shelter. That is the biggest component accounting to about 37% of the ex-gasoline inflation basket. At 3.03% yoy shelter inflation has remained elevated and has countered the decline in yoy core goods inflation from 1.4% at the start of this year to 0.66% in August. However, shelter inflation in recent months has shown signs of decline with the 3m annualized shelter inflation declining to 2.1%. We think this decline will continue because it is more widespread. Looking at shelter inflation at a regional level, the percentage of metro areas that are seeing >4% 3m annualized OER (Owners' equivalent rent) inflation has come down to 26% (Figure 27). More importantly, the cities that are dragging shelter inflation lower will continue to do so in our view. In Figure 28, we show the metro areas with the highest and lowest 3m annualized OER inflation in August. Most of the cities that are currently dragging down shelter inflation are the pandemic boomtowns. Apartment vacancies in those cities are still significantly higher than pre-pandemic levels. So, we expect them to continue pushing overall shelter inflation lower.

Figure 28. The decline in CPI shelter inflation is being driven by continued slowdown in rental growth in the pandemic boom towns.

Metro Area	3m Annualized OER
Riverside	10.70%
St Louis	7.30%
LA - Long Beach	5.90%
Baltimore	5.70%
Alaska	5.00%
New York	4.30%
Miami	0.60%
Seattle	0.40%
Atlanta	-0.30%
Denver	-0.30%
Phoenix	-0.90%
Houston	-1.20%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, BLS, Haver Analytics

Figure 29. ...vacancies in these towns are still elevated and higher than pre-pandemic levels.

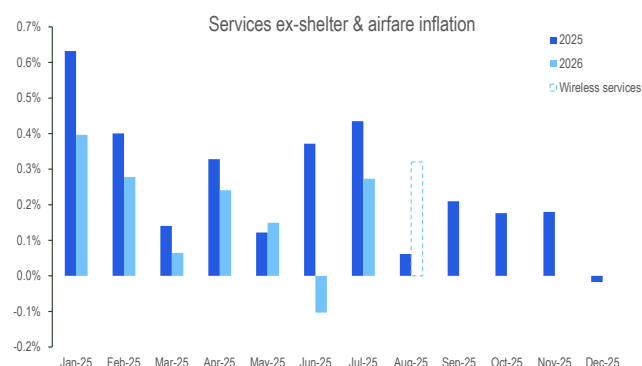


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, ApartmentList, Haver Analytics

The consumer is in a much weaker state than in 2022-23 when producers were able to easily pass down the costs. US savings rate has decreased significantly in the last couple of years. The recent increases in gasoline prices on top of that should lead to further declines in disposable income. This should make consumers more price sensitive and prevent producers from passing down cost increases due to higher energy costs. Both used and new car markets are showing signs of slowing demand in recent months as higher interest rates hit demand. This should cap goods inflation in the near term, which has been running at a 4m annualized pace of 0.31%. That is 25% of the core CPI basket running close to flat.

Services ex-shelter inflation has been gradually declining. If not for the one-time hike in some older plans at a major carrier, we would have had a 0.2% core inflation print. That price increase pushed inflation in the wireless services component by almost 6% MoM in August and contributed to about ~9bp increase in overall core inflation. There seems to be some seasonal factor in core services ex-shelter and airfares inflation. We exclude airfare inflation as it has been a function of oil prices for most of this year. Comparing from month to month vs 2025, we have seen decreases in most months this year (Figure 30). If not for the one-time increase in wireless services component, that would have been the case in August too as inflation in rest of the services was flat in August.

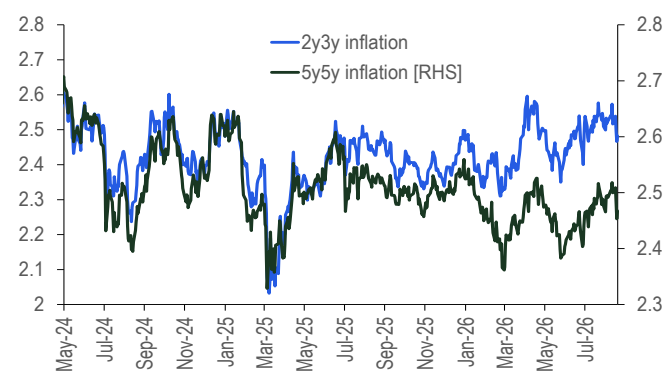
Figure 30. Services ex-shelter and airfares inflation has come in lower compared to last year in most months. That would have been the case in August too if not for the wireless services component.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

Figure 31. Inflation forwards are still trading close to the top of their recent ranges despite the hawkish FOMC.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Inflation expectations are clearly overpriced and will continue to decline. Core goods inflation, which is 25% of core basket, is recently running close to flat and the other two components (shelter and services ex-shelter) are both running below 2.5%. So, it is very hard for ex-gasoline inflation over the next year to get to 2.8%. You need both services and shelter inflation to print close to 3.5% to justify that pricing if goods inflation remains below 1%, so we are getting significant upside buffer in terms of market pricing in those components vs where they are running now. Combined with 1y1y inflation trading close to 2.5%, 2y inflation expectations look overpriced, so we continue to hold our shorts in 2y breakevens hedged with gasoline futures and there is a lot of juice still left in that trade. Inflation forwards such as 2y3y and 5y5y inflation are still trading at the top of their recent ranges despite the hawkish FOMC. So, we see downside risks for breakevens across the curve. For investors who cannot hedge the energy price risk in the front end of the inflation curve, short positions in these forwards are a good option as the beta to oil decreases as we move further along on the inflation curve. *Note: Futures trading involves substantial risk of loss.*

Vol: Cheap front-end bear flattener

Mike Chang

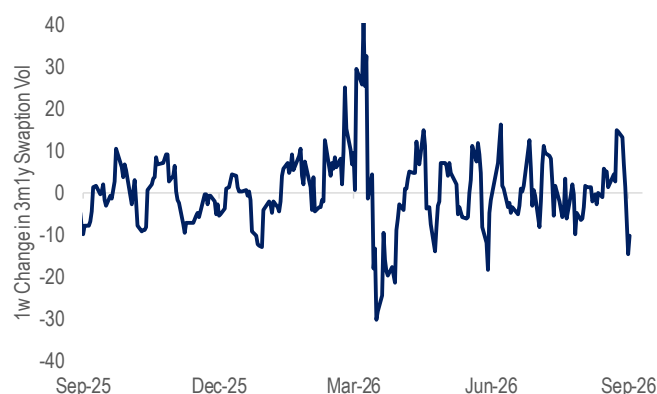
Rates vol has been and should continue to be driven by the expected pace of the Fed's hiking cycle. The 2s5s rate curve could flatten meaningfully in an overly hawkish/policy error scenario, and we recommend buying SFRH7 put against 2y midcurve SFRH9 put as a cheap front-end bear flattener tail hedge. *Note: Futures trading involves substantial risk of loss.*

Front-end tail risks remain asymmetric

As we have previously discussed, it is the pace of the policy rate adjustment that really matters rates vol, and a cautious Fed that is not pressured to act aggressively to beat the forwards should help to keep rates vol anchored even as hikes are realized ([Rates Vol Lab - Slow and steady wins the short gamma race](#)). Even though Fed Chair Warsh was slightly more hawkish than expected, the FOMC press conference does not suggest that this would be the start of an aggressive hiking cycle. This is why swaption vols, especially in the upper-left, have cheapened sharply since the FOMC and week-on-week (Figure 32).

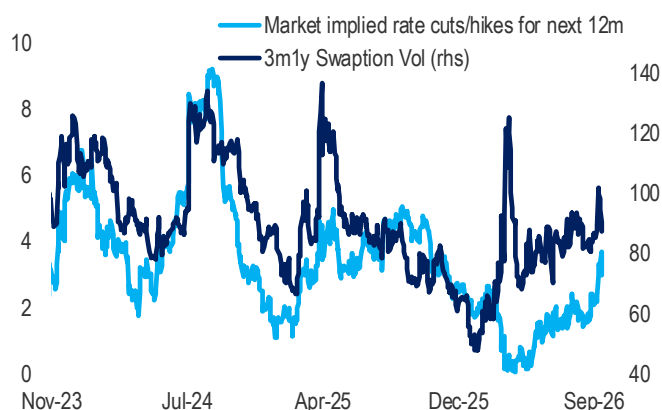
At current level, upper-left vol seems to be close to fair relative to the market's expectation for rate hikes over the next 12 months (Figure 33). Because we believe that the Fed will likely underdeliver relative to the market's current pricing of more than 1 full hike over the next two meetings and 3 hikes in the next 12 months, we see room for short-dated vol to cheapen further from here. On the flip side, if inflation ends up rebounding, which is not our base case, upper-left vol would likely increase as the market reprices for more rate hikes. This strong directionality speaks to the asymmetric risks that we have previously highlighted on front-end rates and should continue to be supportive of upper-left risk reversal skew ([Rates Vol Lab - Trading the asymmetric tail risks](#)). We still recommend "safely" monetizing short-dated vol and upper-left receiver skew through 3m2y receiver ladders.

Figure 32. Upper-left vol cheapened sharply week-on-week despite the rate hike because the Fed has not signaled any deviation from its gradual approach



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Figure 33. The pace of the policy rate adjustment continue will drive rates vol. We are still bearish on gamma given our expectation for a slow and shallow hiking cycle



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

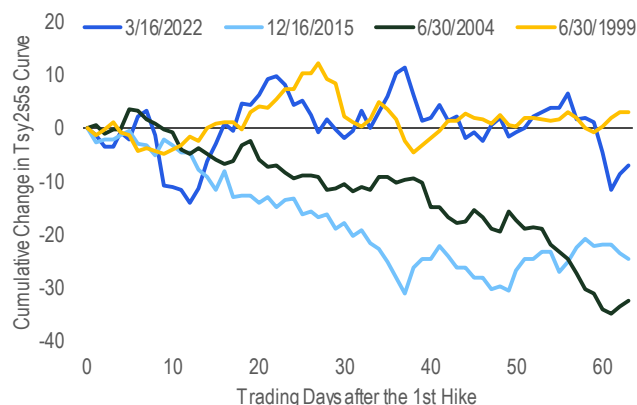
SOFR bear flattener as cheap hawkish hedge

While the direction for rates vol after the first hike tends to be mixed and heavily dependent on the type of hiking cycle that is ultimately carried out, the direction for front-end rates curve has historically leaned to be flatter. Looking at the hiking

cycles since 1999, we see that the 2s5s Tsy curve has flattened 3 out of 4 times 3 months after the first hike by as much as 32bps (Figure 34). The only instance of the 2s5s curve steepening was in 1999 for only 3bps after 3 months.

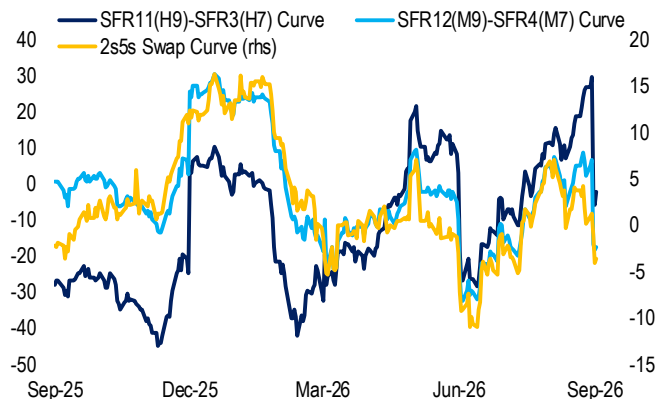
Similarly, we can see front-end rate curve flattening in the current hiking cycle either bullishly with the rate hikes priced out in exchange for delayed rate cuts on gradual inflation improvement or bearishly with the Fed turning even more hawkish and risk causing a policy mistake. For investors looking to hedge for the hawkish Fed/policy mistake curve flattening, 2s5s conditional bear flattener would be an obvious expression but the unfavorable implied vol differential (2y tails higher than 5y tails) would make it costly. Instead, we suggest looking at SOFR futures curves as proxy, where the implied vol differentials are more favorable. Recent history shows that the back whites/back greens SOFR curve has been trading in line with the 2s5s swaps curve (Figure 35), with the generic SOFR curve that corresponds to the current M7/M9 curve looking particularly correlated.

Figure 34. Since 1999, the 2s5s Tsy curve has flattened 3 out of 4 times after the first hike. 1999 was the only exception and 2s5s steepened only 3bps after 3 months.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

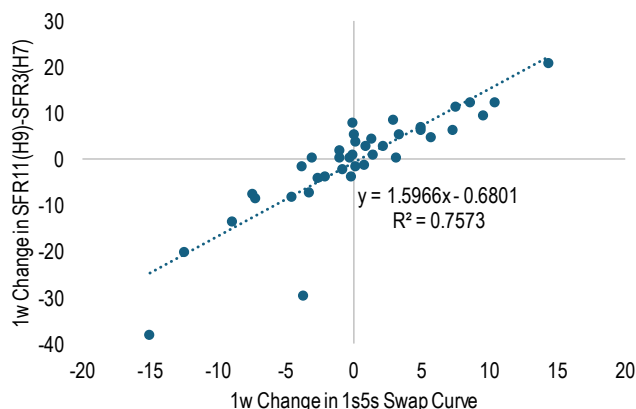
Figure 35. Back whites/back greens SOFR curve has been trading in line with the 2s5s swaps curve, with the M7/M9 curve looking particularly correlated



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

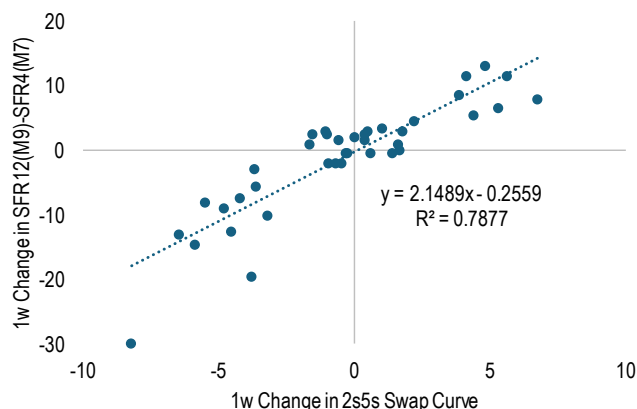
Looking at the non-overlapping weekly changes since the start of the year, we see that changes in the Mar27/Mar29 SOFR curve have actually been more correlated with changes in the 1s5s swap curve (Figure 36). The weekly changes also confirm that the Jun27/Jun29 SOFR curve has been highly correlated with the 2s5s swap curve (Figure 37).

Figure 36. Changes in the Mar27/Mar29 SOFR curve have been more correlated with changes in the 1s5s swap curve



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

Figure 37. Changes in the Jun27/Jun29 SOFR curve have been more correlated with changes in the 2s5s swap curve



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

But despite the relatively high correlation between the moves in the SOFR curve and the 2s5s swap curve, the market is implying very different curve bias based on their implied vol differentials. For example, currently the costless 9m fwd 2s5s bear flattener via swaption payers requires around a 6bp vol give-up relative to the forward, whereas the comparable long SFRM7 put vs short 2y midcurve SFRM9 put would be close to flat. Similarly, the costless 6m fwd 1s5s bear flattener via swaption payers requires around a 2.5bp vol give-up relative to the forward, but the comparable long SFRH7 put vs short 2y midcurve SFRH9 put would result in a very modest premium take-in.

For investors who share our dovish view for a gradual/shallow hiking cycle, we recommend hedging for the hawkish scenario of multiple consecutive hikes by establishing the SFRH7/H9 bear flattener. **Specifically, we recommend buying SFRH7 95.375 put and selling 2QH7 95.4375 put with \$25k/bp risk on each leg at net 6c premium take-in** (pricing as of 2:30pm on 9/18/2026, ref SFRH7=95.425, SFRH9=95.445). Our target/stop for the trade are \$550k/- \$300k respectively. The primary risk to the trade is a persistent bear steepening similar to the one we are experiencing today, but we believe the curve should become more prone to bear flattening as the Fed pricing becomes overly restrictive, i.e., the probability of a policy error should increase going forward on additional tightening.

The conditional bear flattener is also consistent with our revised rates forecasts where we expect the 2s5s and 1s5s curves to flatten 6bps and 9bps respectively by year-end under our base case (see the Overall section for detailed discussion). In a bearish scenario, we believe that the 2s5s and 1s5s curves could flatten 11bps and 14bps respectively. Using the beta from the regression of the weekly changes for SFRH7/H9 and 1s5s swap curve from Figure 36, we estimate that the SFRH7/H9 could flatten by roughly 22bps, which is how we backed into our target for the trade.

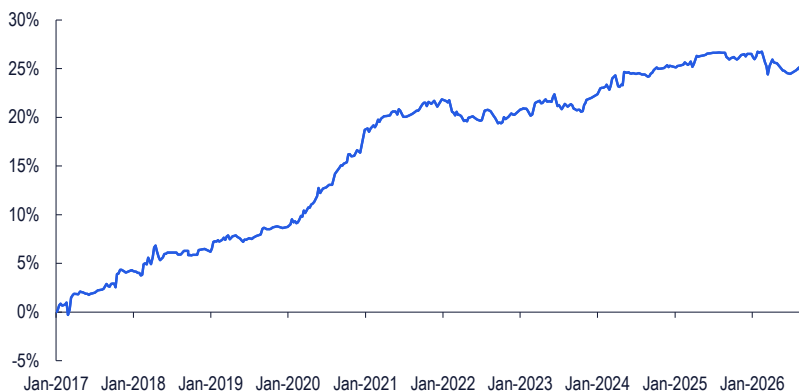
Note: Futures trading involves substantial risk of loss.

US Rates Strategy Model Portfolio Update

The US Rates Strategy Model Portfolio is down 2.98% in 2026.

Figure 38 shows the weekly model portfolio returns since January 2017. Figure 39 shows the P&L from our outstanding trades, while Figure 40 shows all trades closed after 2025. To see the older trades, please refer to a previous publication.

Figure 38. Returns* for the US Rates Strategy Model Portfolio, Jan 2017 – Present



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Past performance is no indicator of future results. Calculations do not include transaction costs and other fees, Note: Returns are reported on a publication-by-publication basis

The portion of this research report regarding non-OCC issued options is not intended for US clients other than Qualified Institutional Buyers. Investing in options is not suitable for all investors. Please see the disclosures concerning the risks of investing in options below and discuss with your Financial Advisor whether this particular options strategy is suitable for you. Note that all option prices are indications, based on pricing as of 4:00pm on September 17, 2026. Interested investors should contact our trading desk for updated price and liquidity information. Also, complex option strategies may entail higher commissions costs.

Outstanding Trade

Long 10y TIPS Opened January 5, 2024; see [US Rates Weekly - The Foxy Five – 2024 edition](#)

1y 5s30s swap steepeners Opened September 5, 2025; see [US Rates Weekly - Trade picking 2025](#)

Long 3y 5s10s30s fly Opened September 26, 2025; see [US Rates Weekly - Fed Funds Shutdown](#)

Long belly 5s10s30s inflation swap Opened November 20, 2025; see [US Rates Strategy - Going long 10y inflation on the curve](#)

Long 5y swap spreads Opened January 8, 2026; see [US Rates Weekly - The Foxy Five – 2026 Edition](#)

Buy 1y CMS 30s/5s curve cap contingent on SPX Opened January 8, 2026; see [US Rates Weekly - The Foxy Five – 2026 Edition](#) (Trade Idea owned by **Mike Chang**)

Long SOFR Z6 vs March 2y SOFR midcurve calls Opened January 12, 2026; see [Investing in 2026: Themes and Trades](#)

Long 1y vs short 3m 30s/5s CMS curve caps Opened January 12, 2026; see [Investing in 2026: Themes and Trades](#)

Pay 1y1y SOFR/TONAR Opened January 16, 2026; see [FX Basis Quarterly - Carrying on but with caution](#)

Costless 1y fwd 10s30s bear steepener via payers Opened January 22, 2026; see [Rates Vol Lab - Vol on 30y can outperform 10y on tail risks](#)

Long 5y TIPS ASW Opened January 30, 2026; see [US Rates Weekly - Loosely neutral, firmly bullish \(the front-end\)](#)

SFRZ6 1x2 put spread Opened March 13, 2026; see [US Rates Weekly - Front-end positioning squeeze](#)

Long 10s20s spreads curve Opened March 27, 2026; see [US Rates Weekly - March showers bring April rallies?](#)

Short SFRU6 Opened April 3, 2026; see [US Rates Weekly - Hedging our bullish lean into CPI](#)

5y IOTA spread widener Opened April 17, 2026; see [US Rates Weekly - Reopening rate cuts?](#)

SOFR Z6/Z7 flattener hedged with long FFQ6 Opened June 17, 2026; see [US Rates Strategy - Widening the aperture](#)

Costless 6m 10s/2s CMS cap ladder Opened June 26, 2026; see [US Rates Weekly - Front-End Flatteners and Long-End Value](#)

Short 2y BEs Opened July 22, 2026; see [US Rates Strategy - Shorting 2y breakevens](#)

Long costless 3m2y receiver ladder Opened July 24, 2026; see [US Rates Weekly - FOMC premiums and Treasury refunding preview](#)

Long 20y Opened August 19, 2026; see [US Rates Strategy - Getting long 20y after today's buyback announcement](#)

Short 3m30y vs long 9m30y payer calendar spread Opened August 21, 2026; see [US Rates Weekly - Can Secretary Bessent and Chair Warsh save the curve?](#)

Long 6m30y payer contingent on SPX Opened September 11, 2026; see [US Rates Weekly - Is a hike the antidote?](#)

Note: Futures trading involves substantial risk of loss.

¹ For trades recommended here, the reasonable basis and risks can be found in the hyperlinked reports as well as the figure in the next page.

Figure 39. Summary of US Rates Strategy Model Portfolio Performance

Trade		Levels	Rationale + Publication Date	
Inflation	Long 10y TIPS, short 3m10y receivers	Level at Open 1.74% Level Current (4:00pm) 2.9% P&L 713 K	We recommend buying 25k DV01 of the July 33 TIPS at 1.74% against \$30mm 3m10y receiver ATMf + 3.52% at 22bps running premium credit. Target is 1% plus received option premium. Stop is 2.25%. Short 3m10y receivers expired on 8/6/25 for profit of 95k. Risk to the trade is a sell-off in real rates on strong activity data. US Rates Weekly - The Fanny Crawl - 2024 Edition	Loss
	Long 10y TIPS, short 3m10y receivers	Target P&L 1.0% Stop P&L 2.25%		
Rates	1y 5s30s swap steepeners hedged with paying Dec FOMC swap	Level at Open 76bp Level Current (4:00pm) 10.8bp P&L -3020 K	Enter 50k dv01 of 1y 5s30s swap steepeners at 75.5bp hedged by paying 25k dv01 of Dec FOMC swap at 3.84% (pricing as of 1:45pm on 9/5/25). We express the trade in swaps as we prefer expressing our view on swap spreads separately (see below), although our spreads view would support a cash 5s30s steepener or receiving in 5y swaps and selling 30y cash. Profit of \$195k taken on pay DEC FOMC swap hedge on 10/31/25. There are two risks to the trade: 1) bear flattening which would be driven by another data "headfake", similar to last year; and 2) a large bull flattening. US Rates Weekly - Trade Journal 2025	Loss
	1y 5s30s swap steepeners hedged with paying Dec FOMC swap	Target P&L 12.7bp Stop P&L 40bp		
Rates	Long 3y forward 5s10s30s fly	Level at Open 27bp Level Current (4:00pm) 13bp P&L 284 K	We are initiating a 20k dv01 long belly/short wings position in 3y forward 5s10s30s fly at 26.8bp (pricing as of 12:30 PM EST on 26th September 2025). Continued flattening of the 10s30s curve is the biggest risk to the trade. US Rates Weekly - End Game: Shortcuts	Profit
	Long 3y forward 5s10s30s fly	Target P&L 12bp Stop P&L 37bp		
Inflation	Long belly 5s10s30s inflation swap	Level at Open 3bp Level Current (4:00pm) 3bp P&L 41 K	We are entering a 40k dv01 long belly position in 5s10s30s inflation swap fly at 2.85bp as of 1:35PM EST on 11/20/2025. A sharp flattening in the 5s10s inflation curve is the biggest risk to the trade. US Rates Weekly - Going long 10y inflation on the curve	Profit
	Long belly 5s10s30s inflation swap	Target P&L 10bp Stop P&L 4bp		
Swap spread	Long 5y swap spreads	Level at Open 27bp Level Current (4:00pm) 27bp P&L 113 K	We buy 50k DV01 of 5y MMS spreads at -26.5bp. Our target is -18bp, and stop-loss for the trade is -34bp. One near-term risk is upcoming financial issuance, although typically belly spreads perform ok at the start of year afterwards. A longer-term risk to the trade is possible foreign reserve manager selling on the back of increased geopolitical. US Rates Weekly - The Fanny Crawl - 2024 Edition	Profit
	Long 5y swap spreads	Target P&L 18bp Stop P&L 24bp		
Rates	Buy 1y CMS 30s1s curve cap contingent on SPX	Level at Open 71bp Level Current (4:00pm) 8bp P&L -150 K	Buy USD 300mm 1y A-25bp 30s1s CMS curve cap contingent on SPX+105p at 5.05c (pricing as of 7:30am 1/8/2026, Refx SPX cash 6910.58, 30s1s CMS Adj ATMf 0.707%). Risk to the trade is a sustained bear flattening in the curve, but the max loss is limited to the premium paid. US Rates Weekly - The Fanny Crawl - 2024 Edition	Loss
	Buy 1y CMS 30s1s curve cap contingent on SPX	Target P&L 95bp Stop P&L 25bp		
Rates	Long SOFR Z6 vs March 2y SOFR midcurve calls	Level at Open 3% Level Current (4:00pm) 4.3163% P&L -2700 K	Buy 1000 SOFR Z6 contracts at 96.825 and sell 1000 March expiry 2y SOFR midcurve calls (underlying is SOFR H8 at 96.635) at a strike of 96.8125 for a 5.5 ktd take-in. Risk to our view include increased fiscal stimulus and growth on the back of AI, combined with somewhat strong inflation prints in Q1. Investing in 2026: Themes and Trades	Loss
	Long SOFR Z6 vs March 2y SOFR midcurve calls	Target P&L 50 ktd Stop P&L 75 ktd		
Volatility	Long 1y vs short 3m 30s1s CMS curve caps	Level at Open -1bp Level Current (4:00pm) 8bp P&L -287 K	Sell 300mm 3m 30s1s CMS cap (0.72%) and buy 300mm 1y 30s1s CMS cap (0.72%) at net premium of \$280k (pricing as of 4pm on 9-Jan-26, 3m ATMf+0.661%, 1y ATMf+0.656%). The primary risk to the trade is a curve flattening in the curve. Investing in 2026: Themes and Trades	Loss
	Long 1y vs short 3m 30s1s CMS curve caps	Target P&L 18bp Stop P&L 28bp		
Front-end	Pay 1y1y SOFR10NAR	Level at Open -31bp Level Current (4:00pm) 20bp P&L 268 K	We now pay 25k dv01 in 1y1y SOFR10NAR at -31bp. The risk to the trade is a VaR shock, perhaps in equity or rates markets or even geopolitical shocks. FX Rates Quarterly - Carrying on but with caution	Profit
	Pay 1y1y SOFR10NAR	Target P&L -23bp Stop P&L -27bp		
Volatility	Costless fly fwd 10s30s bear steepener via payers	Level at Open 26bp Level Current (4:00pm) 4bp P&L -649 K	We recommend selling \$50mm 1y10y 20TM OTM payer and buying \$24mm 1y30y 24TM OTM payer at net zero cost (pricing as of 4:20pm, 1/21/2026: 1y10y atm=3.992%, 1y30y atm=4.247%, DV01=\$40k). The primary risk is a sharp bear flattening on strong growth outlook. Banks Not Safe: Not an 80c coin operation. No, no, no, no, no.	Loss
	Costless fly fwd 10s30s bear steepener via payers	Target P&L 500k Stop P&L 200k		
Inflation	Long 5y TIPS ASW	Level at Open 8bp Level Current (4:00pm) 8bp P&L 195 K	So, we enter a long 20k dv01 position in 5y TIPS asset swap spreads to capture the attractive carry. The biggest risk to this trade is possible tightening of fly nominal swap spreads. US Rates Weekly - Looming real, bond, bullish the horizon	Profit
	Long 5y TIPS ASW	Target P&L 14bp Stop P&L 95bp		
Rates	SFR26 1x2 Put Spread	Level at Open 96.51 Level Current (4:00pm) 7.5163% P&L -125 K	We recommend taking advantage of the recent richening in implied vol and put skew and own a SFR26 96.5 / 96.125 1x2 put spread and collect 2.5c in premium credit to hedge for 1 rate hike this year (pricing as of 1:40pm 3/13/2026, net SFR26=96.505, risk=\$25bp). Note: Futures trading involves substantial risk of loss). The risk is a sharp increase in core inflation later this year. US Rates Weekly - Front-end positioning, squeeze	Loss
	SFR26 1x2 Put Spread	Target P&L 125c Stop P&L 300c		
Rates	Long 10s20s spreads curve	Level at Open -30bp Level Current (4:00pm) 29bp P&L 182 K	We buy \$20k DV01 of 10s20s spreads curve at -30bp with a target of -15bp with a stop at -40bp (Figure 11). The risk to the trade is a continued risk off but we may add onto the trade over time. US Rates Weekly - March shows strong April relief?	Profit
	Long 10s20s spreads curve	Target P&L 15bp Stop P&L 40bp		
Rates	Short SFRU6	Level at Open 3.71% Level Current (4:00pm) 3.69% P&L 744 K	We sell 1000 contracts of SFRU6 at 96.295 as a hedge to our bullish core position into next week's CPI (pricing as of 9am on 4/3/26). Risk to the trade is a weak CPI print but our core long would likely outperform in such a scenario. US Rates Weekly - Helping our bullish bias take off	Profit
	Short SFRU6	Target P&L 25bp Stop P&L 25bp		
Inflation	5y IOTA spread widener	Level at Open 5.8bp Level Current (4:00pm) 3.75bp P&L -73 K	We are entering a 40k dv01 IOTA spread widener in OCS0s at 5bp (pricing as of 10:30AM EST on 4/17/2026) with a target level of 13bp and a stop-loss at 0bps. The biggest risk to this trade is a continued demand for inflation protection leading to breakeven outperforming inflation swaps. US Rates Weekly - Resolving rate cuts?	Loss
	5y IOTA spread widener	Target P&L 13bp Stop P&L 0bp		
Rates	SOFR 2x2/7 flattener hedged with long FFO6	Level at Open -8bp Level Current (4:00pm) 33.5bp P&L -1125 K	We are entering into a SOFR 2x2/7 flattener at 8bps with \$250bp of risk against a long in FF Aug26 future at 96.27 with \$20kbp of risk (pricing as of 4:00pm 6/17/2026). We extend our receiver into the October meeting (\$20kbp of risk of FFH8 3.855% as of 4:35pm on 7/29/26). The risk to the trade is a repricing for a full hiking cycle in 2027. US Rates Weekly - Widening the aperture	Loss
	SOFR 2x2/7 flattener hedged with long FFO6	Target P&L 400k Stop P&L 250k		
Volatility	Costless 6m 10s2s CMS cap ladder	Level at Open 0.077% Level Current (4:00pm) 0.049% P&L 22 K	We recommend buying \$300mm 6m 10s2s ATMf CMS caps and selling \$300mm each of 11bp OTM and 22bp OTM 6m 10s2s CMS caps at net zero cost (pricing as of 1pm on 6/26/2026, CMS adjusted fwd=0.077%, net curve vega=\$5.5k). The risk is a steep steepening in the near-term. US Rates Weekly - Front-end Flatness and Long End Value	Profit
	Costless 6m 10s2s CMS cap ladder	Target P&L 330k Stop P&L 200k		
Inflation	Short 2y BEs hedged with May gasoline futures	Level at Open 2.28% Level Current (4:00pm) 2.3% P&L 472 K	We initiate a 15k dv01 short position in 2y Jul28 breakers at 2.28% (pricing as of 9:15AM EST on Jul 22nd 2026) and hedge it by buying 3k contracts of May27 expiry gasoline futures at 270.70. The biggest risk to the trade is a continued move higher in core inflation expectations. US Rates Weekly - Shorting 2 breakers	Profit
	Short 2y BEs hedged with May gasoline futures	Target P&L 40bp Stop P&L 20bp		
Volatility	Long costless 3m2y receiver ladder	Level at Open 4.20% Level Current (4:00pm) 4.69% P&L 27 K	We recommend buying \$200mm 3m2y ATMf receiver and selling equal notional of 3m2y A-17 and 3m2y A-34 receiver at net zero cost (pricing as of 10:30am 7/24/2026, ATMf+4.20%, net vega=\$5.3k, DV01+\$37.84k, target=\$600k, stop=\$300k). The primary risk to this trade would be an SVB type of steep rate rally. US Rates Weekly - CRR, premiums and Treasury refinance update	Profit
	Long costless 3m2y receiver ladder	Target P&L 500k Stop P&L 300k		
Rates	Long 20y	Level at Open 97.18 Level Current (4:00pm) 5.34% P&L -439 K	We get long \$25mm of May46s at 97.18 (pricing as of 11:30am on 8/19/26). We set the target at 4.9% with a stop at 5.4%. Risk to the trade is a continued buyers strike in the long end of the curve perhaps driven by unanchored inflation expectations. US Rates Weekly - Getting long 20y after today's buyback announcement	Loss
	Long 20y	Target P&L 1.95% Stop P&L 5.40%		
Volatility	Short 3m30y vs long 9m30y payer calendar spread	Level at Open 1.1bp Level Current (4:00pm) 0.8bp P&L 24 K	We recommend selling 20mm 3m30y A-30bp payers and buying 20mm 9m30y A-30A-60 1x1 payer spreads at net 2bps running premium (pricing as of 12:40pm on 8/21/26, 3m30y ATMf+4.575%, 9m30y ATMf+4.588%, net vega=\$2.17k, DV01=\$32k). The trade's primary risk is a sharp sell-off in 30y rate over the next 3 months that exceeds the breakeven of approximately 43bps. US Rates Weekly - Can Secretary Beesett and Chair Walsh save the curve?	Profit
	Short 3m30y vs long 9m30y payer calendar spread	Target P&L 300k Stop P&L 200k		
Volatility	Long 6m30y payer contingent on SPX	Level at Open 4.64% Level Current (4:00pm) 4.64% P&L -7 K	We recommend buying 15Mau27 USD 30mm 30y SOFR ATMf+0.4% payer contingent on SPX < 9% at expiry at 0.49% offer or 3.1bps running premium (pricing as of 11:20am 9/11/2026, net ATMf+4.641%, SPX cash 7668). The primary risks to the trade are a growth slowdown and lower rates. US Rates Weekly - Is a hike the antidote?	Loss
	Long 6m30y payer contingent on SPX	Target P&L 500k Stop P&L 147k		

Pricing as of 4:00pm September 17, 2026. *Units in thousand dollars, unless specified. *This table does not include trade ideas subject to a restriction.

(a) For a detailed list of all closed trades from May 2006 to May 2007, please see "US Rate Strategy — Trade Closeout," US Rate Strategy — Bond Market Roundup: Strategy Citi, May 11, 2007. For a detailed list of all closed trades from May 2007 to May 2008, please see "US Rate Model Portfolio One-Year Anniversary Recap," US Rate Strategy — Bond Market Roundup: Strategy, Citi, May 30, 2008. Between May 2007 and May 2008, the group made a total of 87 trade recommendations, with 50 producing positive results, 36 negative, and one breaking even. This produced a 15.4% total return, with a 1.68 Sharpe ratio. Note: Return on risk is based on Citi's return-on-risk methodology and is calculated by taking the largest two-week change in the trade since January 1997. Return on portfolio based off \$300 million model portfolio sizing. Note: Past performance does not indicate future results. Calculations include transaction fees and other costs, and trades are marked-to-mid. Note: Futures trading involves substantial risk of loss.

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Appendix I: Model Portfolio Closed Trades

Figure 40. US Rates Strategy Closed Trades from 2025 to present

Trade	Inception Date	Unwind Date	Initial	Unwind	P&L (\$000s)	Target P&L	Stop Loss	Risk Return	Portfolio Return
20s30s spreads flattener	Jan 5, 2024	Jan 9, 2025	-0.9bp	-12bp	\$550	-10bp	5bp	55.00%	0.18%
TU net basis	Dec 16, 2024	Feb 7, 2025	-1 tick	0 ticks	\$256			25.60%	0.09%
30y swap spread tightener	Oct 18, 2024	Feb 13, 2025	-81bp	-70bp	(\$261)	-100bp	-70bp	-26.08%	-0.09%
2s10s inflation steepener	Dec 6, 2024	Feb 27, 2025	-12bp	-30bp	(\$500)		-30bp	-50.00%	-0.17%
10y10y vs 20y15y long-dated fwd steepener	Nov 8, 2024	Feb 28, 2025	-94.7bp	-90.2bp	\$173	500k	-300k	17.30%	0.06%
3mf0y 1x2 payer spread	Jan 9, 2025	Feb 28, 2025	4.17%	3.15%	\$52	800k	-300k	5.20%	0.02%
Costless 3m2y 1x2 payer spread	Mar 12, 2025	Apr 4, 2025	3.72%	3.34%	\$42	600k	-250k	4.20%	0.01%
Receive 2y swaps paired with short 3m2y receiver	Jan 9, 2025	Apr 9, 2025	4.06%	3.52%	\$565	760k	-400k	56.54%	0.19%
Long 3y swap spreads	Apr 3, 2025	Jun 6, 2025	-24.5bp	-29bp	(\$225)	-20bp	-28bp	-22.50%	-0.08%
Long 5y5y real rates vs long 5y5y inflation	Mar 21, 2025	Apr 10, 2025	91bp	-20bp	(\$300)	60bp	-20bp	-30.00%	-0.10%
Short 10y UST vs long 10y Bund	Mar 7, 2025	Apr 11, 2025	142bp	200bp	\$3,039	190bp	125bp	303.90%	1.01%
Long 2y1y inflation hedged with SOFR puts	Apr 23, 2025	May 9, 2025	2.18%	2.25%	\$260	30bp	-20bp	26.00%	0.09%
TU net basis	Feb 21, 2025	May 16, 2025	-4.0	-2.8	\$60	3bp	-2bp	6.00%	0.02%
Long CORM5	May 8, 2025	Jun 6, 2025	97.4	97.3	(\$200)	98	97	-20.00%	-0.07%
Conditional SOFR M5/M6 steepeners	Jan 9, 2025	Jun 13, 2025	-8 ticks	0 ticks	\$0	+25 ticks	-15 ticks	0.00%	0.00%
5y IOTA spread tightener	Jan 9, 2025	Jun 27, 2025	15.5bp	7.5bp	\$280	3bp	22bp	28.00%	0.09%
Long belly of 10s20s30s fly	Aug 9, 2024	Jun 27, 2025	46bp	59.5bp	(\$166)	30bp	55bp	-16.63%	-0.06%
3m2y 1x2 payer spread	Apr 11, 2025	Jun 27, 2025	3.57%	3.50%	\$115	700k	-300k	11.50%	0.04%
Short 3m6y straddle	Apr 23, 2025	Jun 27, 2025	3.62%	3.45%	\$705	500k	-250k	70.50%	0.24%
3mfly SOFR/TONAR	Dec 13, 2024	Jul 25, 2025	-38bp	-29.5bp	\$393	-48bp	-32bp	39.30%	0.13%
Costless 6m 5s30s bear steepener	May 16, 2025	Aug 22, 2025	36.5bp	-67.9bp	\$92	500k	-300k	9.20%	0.03%
3m2y 1x2 payer spread	Jun 27, 2025	Aug 22, 2025	3.50%	3.45%	\$120	700k	-300k	12.00%	0.04%
Short 2y2y inflation	Jun 27, 2025	Aug 26, 2025	2.34%	2.48%	(\$445)	2.15%	2.46%	-44.50%	-0.15%
Long 3mf0y straddle	Jul 25, 2025	Sep 12, 2025	88bp	75.8bp	(\$175)	105bp	80bp	-17.50%	-0.06%
Short 1y inflation with 5y5y hedge	Sep 5, 2025	Oct 29, 2025	-80bp	-55bp	\$243	-50bp	-100bp	24.30%	0.08%
3mfly 1x1 payer spread	Oct 17, 2025	Oct 31, 2025	3.19%	3.45%	\$195	720k	-330k	19.50%	0.07%
Long 2m80y straddle vs short 1m80y straddle	Nov 7, 2025	Nov 26, 2025	2bp	12bp	\$32	250k	-150k	3.20%	0.01%
Long belly 10s20s30s cash fly	Oct 17, 2025	Dec 5, 2025	53.5bp	60bp	(\$190)	45bp	60bp	-19.00%	-0.06%
Costless bullish 2s5s10s conditional fly	Sep 5, 2025	Dec 5, 2025	-31.5bp	-21.5bp	\$0	-51.5bp	-18.5bp	0.00%	0.00%
Conditional Z5/Z6 bull steepener	Apr 4, 2025	Dec 18, 2025	-31.5bp	0.87bp	\$193	500k	-300k	19.30%	0.06%
Long SOFR Z6 vs short Dec expiry 2y mdc curve calls	Sep 5, 2025	Jan 8, 2026	2.9%	3.12%	(\$194)	45 ticks	-30 ticks	-19.40%	-0.06%
Long 2y swap spreads with a short SOFR/FF Z5 hedge	Sep 5, 2025	Jan 8, 2026	-23.2 bp	-16.2bp	\$600	-18.2bp	-28.2bp	60.00%	0.20%
Long 10y10y vol	Jul 11, 2025	Jan 8, 2026	1.77bp vol	1.04bp vol	(\$525)	500k	-375k	-52.50%	-0.18%
Costless 3mfly 1x2 receiver spread	Nov 21, 2025	Jan 23, 2026	3.32%	3.50%	\$16	700k	-300k	1.60%	0.01%
Weekly Straddle on TYH6	Feb 5, 2026	Feb 12, 2026	38 mini ticks	8 mini ticks	\$100	700k	-475k	10.00%	0.03%
Long 1y10y straddle vs short 3mf0y straddle	Oct 17, 2025	Feb 20, 2026	4.11%	3.76%	(\$351)	500k	-350k	-35.10%	-0.12%
Short March CPI	Feb 27, 2026	Mar 5, 2026	328	329	(\$94)	328	329	-9.40%	-0.03%
3mfly OTM Payer	Feb 27, 2026	Mar 5, 2026	3	3	\$262	300k	-143k	26.20%	0.09%
1x2 put spreads on SFRH7	Jan 8, 2026	Mar 13, 2026	3.12%	-3.75 ticks	(\$94)	12.5 ticks	-8 ticks	-9.38%	-0.03%
20s30s swap spread flattener	Apr 17, 2025	Mar 13, 2026	-5.6bp	-4bp	(\$214)	-15bp	0bp	-21.40%	-0.07%
Costless 3mfly Receiver Ladder	Mar 6, 2026	Mar 20, 2026	3.37%	3.86%	\$33	600k	-250k	3.30%	0.01%
6m ATM 30s/2s CMS cap vs 6mfly ATM payer	Jan 30, 2026	Mar 20, 2026	-277bp	-318bp	(\$380)	450k	-300k	-38.00%	-0.13%
2y-3y2y inflation swap steepener hedged with Sep expiry WTI f	Jan 8, 2026	Mar 27, 2026	2bp	-37bp	(\$270)	30bp	-20bp	-27.00%	-0.09%
Long 1y5y straddle vs short 3m6y straddle	Jan 12, 2026	Apr 2, 2026	7.5bp	3.62%	\$693	500k	-250k	69.30%	0.23%
Long SERFFJ6	Feb 27, 2026	Apr 9, 2026	-2.5bp	0.5bp	\$208	1bp	-5bp	20.80%	0.07%
Receive 1y2y SOFR on a covered basis	Jan 8, 2026	May 15, 2026	3.28%	3.89%	(\$1,749)	1000k	1400k	-174.90%	-0.58%
Long 9mfly vs short 3mfly receiver calendar spread	Mar 27, 2026	May 22, 2026	-6.3bp	12.1bp	\$197	600k	300k	19.70%	0.07%
Long 30y BE	Apr 1, 2026	May 29, 2026	2.2%	2.27%	\$340	20bp	2%	34.00%	0.11%
May Fed Fund Futures (FFK6)	Feb 5, 2026	Jun 1, 2026	96.46	96.37	(\$183)	10	10	-18.30%	-0.06%
5s30s calendar spread steepener	Jun 6, 2025	Jun 8, 2026	44bp	34bp	(\$371)	600k	-350k	-37.10%	-0.12%
SFRZ6 Risk Reversal	May 29, 2026	Jun 17, 2026	4 ticks	-17 ticks	(\$103)	40 ticks	20 ticks	-10.30%	-0.03%
3m 30s/2s CMS straddles vs curve cap/floor strangles	Apr 2, 2026	Jun 26, 2026	0.53%	0.17%	\$290	645k	-300k	29.00%	0.10%
Long 1x2 call spreads on SFRH6	Nov 7, 2025	Apr 1, 2026	1 tick	0 ticks	\$25	500k	-300k	2.50%	0.01%
Short 1mf0y ATM straddles	Jun 17, 2026	Jul 10, 2026	4.08%	4.13%	\$190	475k	-250k	19.00%	0.06%
Short dec '26 CPI vs XBZ6	Jun 25, 2026	Jul 14, 2026	335.05	334.80	\$180	450k	-300k	18.00%	0.06%
Long TY straddles	Jul 10, 2026	Jul 15, 2026	30 ticks	9.5 ticks	(\$69)	350k	-187.5k	-6.90%	-0.02%
Long 1y1y Real Rates	Mar 13, 2026	Jul 24, 2026	0.97%	1.82%	(\$1,258)	0.65%	1.25%	-125.80%	-0.42%
Long TY straddles	Aug 7, 2026	Aug 14, 2026	38 ticks	108-18	(\$169)	475k	-237.5k	-16.90%	-0.06%
US/EUR 5y5y inflation swap spread widener	Jun 11, 2026	Aug 20, 2026	22.5bp	33bp	\$220	45bp	12bp	22.00%	0.07%
Costless 3m2y 1x2 payer spread	May 22, 2026	Aug 21, 2026	3.94%	4.09%	\$430	600k	300k	43.00%	0.14%
Short SERFFQ6	Jun 26, 2026	Aug 31, 2026	-1.5bp	-1.13bp	(\$15)	-4bp	-1bp	-1.54%	-0.01%
FFN6/FFQ6 Flattener	Mar 5, 2026	Sep 1, 2026	-6 ticks	0.2 ticks	(\$408)	0 ticks	-1 tick	-40.80%	-0.14%
Costless 1y fwd 5s30s conditional bull steepener	Sep 5, 2025	Sep 7, 2026	77.5bp	41.9bp	\$0	122.5bp	52.5bp	0.00%	0.00%
Long 9m2y vs short 3m2y receiver calendar spread	May 22, 2026	Sep 11, 2026	1.5bp	4.62%	\$75	500k	-250k	7.50%	0.03%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Appendix II: Treasury Relative Value

Figure 41. Richest and cheapest treasury notes & bonds by sector

		Spread to				Asset Swap					
		Curve (bp)	3m Median Spread (bp)	3m Percentile			Spread (bp)	3m Median ASW (bp)	3m Percentile		
1y-2y	Cheap	3.375% Dec27	2.73	0.10	100%	1y-2y	Cheap	3.875% Jul28	9.9	12.3	8%
		3.875% Nov27	2.56	0.55	100%			3.375% Sep28	11.6	14.0	6%
		3.375% Feb28	0.81	-0.31	100%			3.375% Feb28	7.8	10.3	5%
		0.375% Sep27	-1.82	0.31	0%			3.5% Sep27	-1.4	7.5	2%
		3.5% Sep27	-2.50	0.19	0%			0.375% Sep27	-1.5	7.4	0%
	Rich	3.875% Oct27	-3.47	0.22	0%		Rich	3.875% Oct27	-2.6	8.1	0%
2y-3y	Cheap	4% Jul29	0.70	0.19	98%	2y-3y	Cheap	4.125% Mar29	16.4	18.1	13%
		3.125% Aug29	0.51	0.08	74%			3.125% Aug29	17.4	19.1	14%
		3.5% Mar29	0.25	0.03	82%			4% Jul29	17.3	19.1	16%
		1.625% Aug29	-2.35	-1.51	7%			4.375% Nov28	12.4	15.4	5%
		3.125% Nov28	-0.44	0.41	0%			1.375% Oct28	11.2	14.3	5%
	Rich	1.25% Sep28	-1.18	-0.26	0%		Rich	3.125% Nov28	11.4	14.6	3%
3y-5y	Cheap	3.5% Feb31	0.06	-0.49	100%	3y-5y	Cheap	3.5% Feb31	24.3	25.3	33%
		4.25% Jun31	0.50	0.10	100%			0.875% Nov30	22.5	23.5	35%
		0.875% Nov30	0.65	0.26	98%			3.625% Sep30	23.2	24.3	29%
		3.875% Apr30	0.04	0.49	0%			4.125% Nov29	18.9	21.1	10%
		4% Mar30	-0.40	0.20	2%			4% Mar30	21.1	23.4	10%
	Rich	3.625% Mar30	0.03	0.70	0%		Rich	4% Oct29	18.3	20.6	10%
5y-7y	Cheap	4.5% Dec31	-0.03	-0.40	91%	5y-7y	Cheap	1.375% Nov31	25.0	26.7	14%
		1.375% Nov31	0.35	0.04	84%			1.875% Feb32	26.0	27.9	19%
		3.625% Sep31	-0.15	-0.37	82%			4.5% Dec31	26.9	28.9	16%
		3.875% Sep32	-0.10	0.21	0%			3.75% Nov32	29.3	31.8	11%
		3.75% Feb33	-0.38	-0.06	0%			3.75% Oct32	29.5	32.0	14%
	Rich	3.75% Oct32	-0.32	0.05	0%		Rich	4.125% Nov32	29.2	31.9	13%
7y-10y	Cheap	4.25% Aug35	0.73	0.30	96%	7y-10y	Cheap	4.5% Nov33	33.2	35.3	22%
		4% Nov35	0.19	-0.21	86%			4% Feb34	33.2	35.4	17%
		4.5% Nov33	0.72	0.49	100%			3.875% Aug34	34.0	36.3	13%
		3.875% Aug34	0.59	0.72	14%			4% Nov35	35.2	38.0	8%
		4.25% Nov34	0.56	0.78	2%			4.125% Feb36	35.1	38.2	8%
	Rich	4.375% May34	0.59	0.85	7%		Rich	4.5% Feb36	25.9	29.4	6%
10y-20y	Cheap	3.5% Feb39	-7.59	-8.27	100%	10y-20y	Cheap	4.75% Feb37	29.4	34.2	6%
		4.25% May39	-3.88	-4.20	79%			5% May37	29.3	34.3	6%
		4.625% Feb46	-0.30	-0.58	98%			3.5% Feb39	32.1	37.7	3%
		1.125% Aug40	-1.58	-0.60	0%			4.75% Feb41	46.1	54.4	2%
		1.75% Aug41	-1.13	-0.08	0%			3.875% Aug40	41.7	50.5	2%
	Rich	1.125% May40	-1.83	-0.73	11%		Rich	4.25% Nov40	43.5	52.5	2%
20y+	Cheap	4.75% Feb56	-0.16	-0.49	100%	20y+	Cheap	1.375% Aug50	41.6	47.2	2%
		4.625% Nov55	0.14	-0.15	100%			1.875% Nov51	44.7	50.3	2%
		4.75% Aug55	0.16	-0.05	91%			1.625% Nov50	43.4	49.0	2%
		3% May47	-1.35	-1.00	19%			4% Nov52	57.1	64.1	2%
		2% Feb50	-2.02	-1.66	18%			4.625% May54	60.2	67.4	2%
	Rich	1.25% May50	-4.59	-3.88	21%		Rich	4.75% Nov53	61.2	68.8	2%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Appendix III: Swap Fly Relative Value

Figure 42. Equal weighted fly relative value table

9/17/2026		1:2:1 SOFR swap					6/17/2026					
Fly Structure	Fly Structure	Weights	Level (bp)	3m C+R (bp)	Z Score			3m High (bp)	3m Low (bp)	3m Correlation		
					1m ZS	3m ZS	6m ZS			10s	2s 10s	5s30s
2y, 3y, 5y	2-3-5	-0.5:1-0.5	15	-0.9	2.10	3.50	2.11	15	-14	7%	-65%	-66%
2y, 5y, 7y	2-5-7	-0.5:1-0.5	-2.0	-2.6	-142	0.46	0.78	0.3	-6.9	47%	79%	44%
3y, 5y, 7y	3-5-7	-0.5:1-0.5	-16	-0.8	-0.44	0.85	1.26	-0.8	-3.8	53%	54%	14%
2y, 5y, 10y	2-5-10	-0.5:1-0.5	-3.6	-2.6	100	189	2.08	-2.0	-110	65%	21%	-24%
2y, 10y, 30y	2-10-30	-0.5:1-0.5	-3.8	-2.3	-3.03	-0.35	-0.75	10	-8.4	49%	87%	53%
3y, 5y, 10y	3-5-10	-0.5:1-0.5	-3.2	-0.7	163	2.89	2.77	-3.0	-8.2	48%	-52%	-83%
2y, 7y, 10y	2-7-10	-0.5:1-0.5	-4.0	-2.9	-2.96	-0.44	-0.75	10	-8.0	36%	93%	66%
3y, 7y, 10y	3-7-10	-0.5:1-0.5	-3.7	-1.1	-3.07	-1.06	-1.57	0.0	-4.8	34%	93%	68%
5y, 7y, 10y	5-7-10	-0.5:1-0.5	-1.8	-0.1	106	2.14	2.44	-1.5	-2.6	70%	11%	-25%
5y, 7y, 12y	5-7-12	-0.5:1-0.5	-3.7	-0.1	168	2.81	2.94	-3.5	-6.5	50%	-66%	-91%
2y, 10y, 15y	2-10-15	-0.5:1-0.5	-5.5	-2.9	-2.91	-1.29	-1.88	4.2	-7.2	24%	99%	79%
3y, 10y, 15y	3-10-15	-0.5:1-0.5	-5.1	-1.1	-2.71	-2.11	-2.34	3.3	-5.1	18%	99%	84%
3y, 10y, 20y	3-10-20	-0.5:1-0.5	-7.3	-0.8	-3.16	-1.72	-2.04	-1.3	-7.6	31%	95%	72%
5y, 10y, 20y	5-10-20	-0.5:1-0.5	-5.4	0.1	-3.61	-2.34	-2.03	-3.5	-5.4	40%	81%	55%
5y, 10y, 30y	5-10-30	-0.5:1-0.5	-1.6	0.5	0.26	1.06	0.37	-0.2	-3.1	72%	18%	-22%
7y, 10y, 15y	7-10-15	-0.5:1-0.5	-3.0	0.0	-0.95	0.10	-0.35	-2.6	-3.3	48%	26%	-2%
7y, 10y, 20y	7-10-20	-0.5:1-0.5	-5.1	0.3	132	2.21	2.51	-4.7	-7.8	54%	-45%	-76%
7y, 10y, 30y	7-10-30	-0.5:1-0.5	-1.3	0.6	151	2.41	2.74	-0.9	-6.8	51%	-63%	-89%
10y, 12y, 15y	10-12-15	-0.5:1-0.5	-0.8	0.1	158	2.60	2.86	-0.8	-1.1	48%	-22%	-42%
5y, 15y, 20y	5-15-20	-0.5:1-0.5	3.8	0.0	-2.17	-2.96	-2.71	15.3	3.8	-12%	94%	98%
5y, 15y, 30y	5-15-30	-0.5:1-0.5	7.6	0.3	-2.36	-3.05	-2.55	16.3	7.6	0%	96%	93%
7y, 15y, 20y	7-15-20	-0.5:1-0.5	4.0	0.1	-2.10	-3.02	-2.76	11.3	4.0	-17%	91%	98%
7y, 15y, 30y	7-15-30	-0.5:1-0.5	7.8	0.5	-2.43	-3.26	-2.47	12.2	7.8	2%	93%	90%
10y, 15y, 20y	10-15-20	-0.5:1-0.5	2.4	0.2	-2.14	-3.09	-2.64	4.9	2.4	-15%	87%	90%
10y, 15y, 30y	10-15-30	-0.5:1-0.5	6.3	0.5	0.51	0.73	-0.19	6.7	5.8	64%	-2%	-37%
5y, 20y, 30y	5-20-30	-0.5:1-0.5	11.9	-0.2	-2.06	-2.87	-2.68	25.6	11.9	-16%	91%	99%
7y, 20y, 30y	7-20-30	-0.5:1-0.5	12.1	0.0	-1.97	-2.87	-2.71	21.5	12.1	-23%	87%	99%
10y, 20y, 30y	10-20-30	-0.5:1-0.5	10.5	0.0	-1.85	-2.75	-2.59	15.1	10.5	-27%	78%	94%
12y, 20y, 30y	12-20-30	-0.5:1-0.5	8.7	0.0	-1.79	-2.67	-2.44	10.9	8.6	-23%	61%	78%
15y, 20y, 30y	15-20-30	-0.5:1-0.5	6.0	0.1	123	1.78	1.08	6.1	5.4	19%	-42%	-44%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Notes: Level, carry, Z-scores and correlations are for positions receiving the belly. Positive number indicate cheapness in the belly, negative numbers indicate richness. Levels are for a -0.5:1-0.5 weighted fly. Correlations are calculated over the most recent 3m window.

Figure 43. PCA weighted fly relative value table

9/17/2026		PCA SOFR swap					6/17/2026					
Fly Structure	Fly Structure	Weights	Level (bp)	3m C+R (bp)	Z Score			3m High (bp)	3m Low (bp)	3m Correlation		
					1m ZS	3m ZS	6m ZS			10s	2s 10s	5s30s
2y, 3y, 5y	2-3-5	-0.52:1-0.47	7.6	-1.0	193	3.20	2.17	7.6	4.0	29%	-56%	-68%
2y, 5y, 7y	2-5-7	-0.22:1-0.76	7.0	-0.9	170	2.95	2.49	7.1	15	52%	-55%	-85%
3y, 5y, 7y	3-5-7	-0.36:1-0.63	3.4	-0.4	157	2.76	2.60	3.5	0.1	54%	-43%	-76%
2y, 5y, 10y	2-5-10	-0.35:1-0.61	14.4	-1.6	163	2.86	2.56	14.9	4.0	62%	-47%	-81%
2y, 10y, 30y	2-10-30	-0.28:1-0.66	21.6	-0.7	104	2.06	2.39	24.3	11.3	86%	4%	-43%
3y, 5y, 10y	3-5-10	-0.52:1-0.46	6.9	-0.7	152	2.70	2.68	7.4	0.9	65%	-36%	-73%
2y, 7y, 10y	2-7-10	-0.17:1-0.81	8.0	-0.9	160	2.84	2.67	8.4	17	67%	-43%	-78%
3y, 7y, 10y	3-7-10	-0.25:1-0.74	4.5	-0.5	150	2.70	2.80	5.0	0.3	72%	-32%	-70%
5y, 7y, 10y	5-7-10	-0.48:1-0.51	15	-0.1	136	2.52	2.92	18	0.2	79%	-10%	-48%
5y, 7y, 12y	5-7-12	-0.57:1-0.41	18	-0.2	123	2.35	2.74	2.3	0.0	84%	-7%	-48%
2y, 10y, 15y	2-10-15	-0.13:1-0.84	7.4	-0.6	138	2.54	2.57	8.2	10	79%	-28%	-69%
3y, 10y, 15y	3-10-15	-0.18:1-0.79	4.0	-0.2	123	2.35	2.60	4.8	-0.4	82%	-16%	-58%
3y, 10y, 20y	3-10-20	-0.27:1-0.69	6.6	-0.1	0.95	2.00	2.40	8.3	10	84%	8%	-38%
5y, 10y, 20y	5-10-20	-0.43:1-0.55	2.1	0.2	0.16	1.16	1.24	3.4	0.1	76%	38%	-2%
5y, 10y, 30y	5-10-30	-0.54:1-0.45	5.1	0.4	-134	-0.04	-0.64	6.6	3.6	68%	55%	19%
7y, 10y, 15y	7-10-15	-0.49:1-0.50	0.2	0.1	0.28	1.31	1.59	0.6	-0.5	66%	17%	-16%
7y, 10y, 20y	7-10-20	-0.61:1-0.38	0.4	0.2	-151	-0.16	-0.65	1.1	-0.1	56%	51%	19%
7y, 10y, 30y	7-10-30	-0.71:1-0.28	2.0	0.3	-3.57	-2.54	-1.93	3.1	2.0	44%	67%	41%
10y, 12y, 15y	10-12-15	-0.53:1-0.47	0.5	0.1	-0.22	0.83	0.60	0.5	0.3	47%	49%	32%
5y, 15y, 20y	5-15-20	-0.17:1-0.82	1.1	0.3	-2.37	-1.41	-1.49	2.0	0.8	43%	66%	39%
5y, 15y, 30y	5-15-30	-0.34:1-0.67	5.4	0.6	-3.57	-3.27	-1.92	7.9	5.4	39%	81%	58%
7y, 15y, 20y	7-15-20	-0.25:1-0.75	0.4	0.3	-2.97	-3.40	-2.18	1.5	0.4	22%	72%	54%
7y, 15y, 30y	7-15-30	-0.44:1-0.56	3.4	0.6	-2.73	-3.90	-2.27	6.2	3.4	10%	87%	77%
10y, 15y, 20y	10-15-20	-0.40:1-0.60	0.2	0.2	-2.46	-3.61	-2.34	1.2	0.2	1%	71%	63%
10y, 15y, 30y	10-15-30	-0.63:1-0.38	2.2	0.3	-2.25	-3.52	-2.34	4.4	2.2	-18%	85%	89%
5y, 20y, 30y	5-20-30	-0.20:1-0.81	5.2	0.3	-2.70	-3.66	-2.03	7.2	5.2	20%	70%	60%
7y, 20y, 30y	7-20-30	-0.26:1-0.75	4.0	0.3	-2.20	-3.53	-2.24	6.2	4.0	-9%	66%	70%
10y, 20y, 30y	10-20-30	-0.37:1-0.65	3.3	0.2	-1.84	-3.12	-2.26	5.2	3.2	-33%	48%	66%
12y, 20y, 30y	12-20-30	-0.45:1-0.56	2.5	0.1	-1.67	-2.94	-2.29	4.1	2.4	-38%	28%	49%
15y, 20y, 30y	15-20-30	-0.59:1-0.42	1.9	0.0	-1.13	-2.28	-2.08	2.6	1.6	-27%	-7%	12%

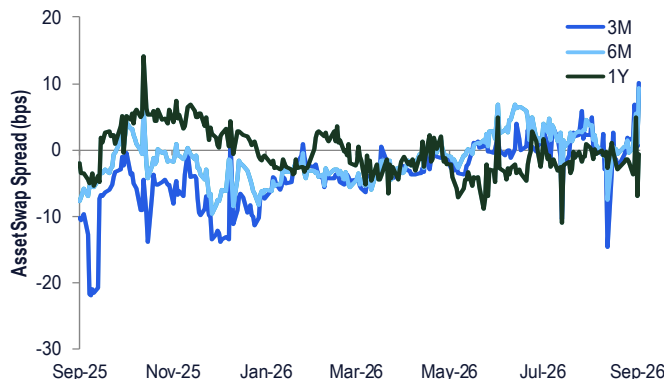
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Notes: Level, carry, Z-scores and correlations are for positions receiving the belly. Positive number indicate cheapness in the belly, negative numbers indicate richness. PCA weights are computed over a 5-year rolling window. PCA weighting is based on the weights of the 3rd principal component of the three legs of the fly over the analysis window. This reduces exposure to parallel (1st PC) and curve (2nd PC) moves among the three legs, but does not eliminate exposure or correlation to all parts of the yield curve at all times. Correlations are calculated over the most recent 3m window.

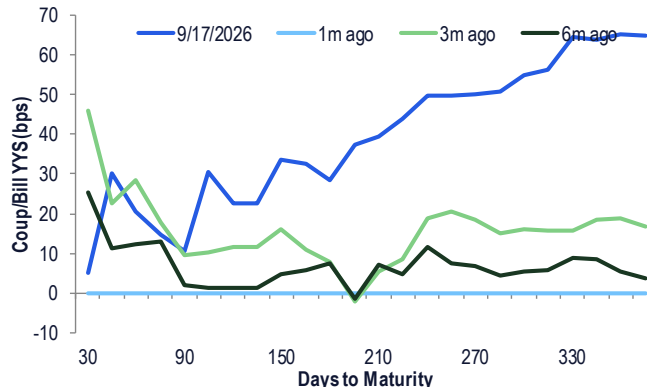
Appendix IV: Front End Monitor

Figure 44. Historical time series of ASW of OTR 3m, 6m and 1y bills



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Figure 45. Term Structure of Coup/Bill Yield Spread at different times



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Figure 46. SOFR curve monitor

Curves	9/17/2026	Z-scores			Regression*		
		1m	6m	12m	Correlation	Beta	Residual
SOFR Futures							
Z6/Z7	36.5	1.4	2.4	2.5	16%	0.05	28.4 (2.1)
Z7/Z8	-13.5	-1.9	-1.7	-1.9	-83%	-0.31	6.3 (0.9)
Z8/Z29	-5.5	-2.0	-2.3	-2.6	-86%	-0.22	-0.8 (-0.2)
Z29/Z30	2.0	-1.8	-2.3	-2.9	-85%	-0.19	-2.4 (-0.9)
Z30/Z31	6.0	-2.2	-3.3	-3.8	-77%	-0.13	-5.9 (-2.9)
SOFR Packs							
White/Red	13.7	1.1	2.2	1.0	-32%	-0.09	22.4 (2)
Red/Green	-10.4	-1.6	-1.5	-1.7	-82%	-0.29	7.4 (1.2)
Green/Blue	-5.9	-2.2	-2.5	-2.8	-86%	-0.22	-1.9 (-0.6)
Blue/Gold	-109.9	-1.6	-2.5	-3.0	-96%	-0.60	-3.5 (-1.3)
Constant Maturity SOFR							
1y/2y	-10.0	-1.8	0.4	-0.2	-22%	-0.05	2.8 (0.3)
2y/3y	-10.1	-2.0	-2.0	-2.0	-75%	-0.29	-1.7 (-0.2)
3y/4y	0.7	-1.8	-1.9	-1.9	-89%	-0.24	3.1 (1)
4y/5y	5.8	-1.6	-2.7	-3.5	-53%	-0.03	-3.1 (-2.6)

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, *Notes: 1. Regression of each curve is against the longer leg of the curve. -ve residual value suggests that the curve is too flat and +ve suggests it is too steep. 2. Constant Maturity Eurodollars are computed using linear interpolation between Eurodollar futures. Note: Futures trading involves substantial risk of loss.

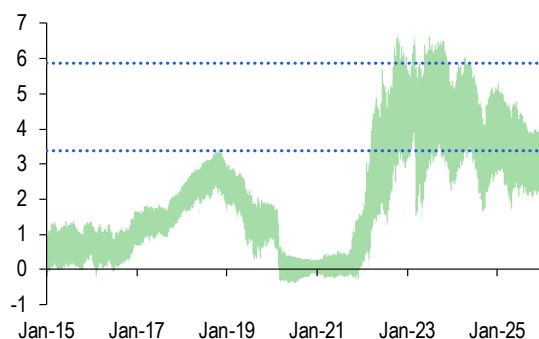
Figure 47. SOFR fly monitor

Flies	9/17/2026	1:2:1 Butterflies				PCA Butterflies		
		1y Z-score	Correlation	Beta	Residual	1y Z-Score	Residual	Weights
SOFR Futures								
Z6/Z7/Z8	50.0	3.4	79%	0.31	21.4 (2.1)	0.7	4.1	0.6:2:1.9
Z7/Z8/Z29	-8.0	-0.8	-65%	-0.12	6.1 (1.4)	2.0	3.4	0.6:2:1.5
Z8/Z29/Z30	-7.5	-1.5	-63%	-0.06	0.6 (0.3)	1.0	1.2	0.8:2:1.3
Z29/Z30/Z31	-4.0	-0.8	-73%	-0.07	2.4 (1.8)	3.3	3.6	0.8:2:1.2
SOFR Packs								
W/R/G	24.1	3.2	62%	0.15	14.7 (1.9)	0.5	1.8	0.7:2:1.7
R/G/B	-4.5	-0.2	-61%	-0.10	8.1 (2)	3.5	5.6	0.6:2:1.5
G/B/Go	104.0	2.5	90%	0.16	-0.3 (-0.1)	1.0	1.2	0.8:2:1.7
Constant Maturity SOFR								
1y/2y/3y	0.1	2.1	81%	0.21	0.9 (0.1)	0.1	0.0	0.9:2:1.4
2y/3y/4y	-10.8	-1.6	-38%	-0.09	-4.8 (-0.7)	-1.5	-6.5	0.4:2:1.8
3y/4y/5y	-5.0	-1.3	-84%	-0.20	6 (1.8)	2.7	5.9	0.2:2:1.9

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, W=White, R=Red, G=Green, B=Blue, Go =Gold and P=Purple. Notes: 1. Correlation, beta and residuals are computed by regressing the fly against its belly. -ve residual value suggests that the fly is rich, +ve suggests that it is cheap. 3. Note that PCA butterflies are created such that there is no correlation and beta with belly of the fly. Note: Futures trading involves substantial risk of loss.

Figure 48. Market-implied 1y fwd FF +/- 1SD



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Haver Analytics, Federal Reserve Bank of Kansas City

Appendix V: FX Hedged Yield Pickup

Figure 49. FX hedged yield pickup in the 2y sector using 3m forward hedging (annualized)

swapped ccy/asset ccy	*USTs	*Bunds	*Gilts	*JGBs	*AGBs	*CGBs
USD	0	6 (-0.4)	25 (0.4)	10 (-1.3)	-24 (-0.4)	37 (-0.2)
EUR	-6 (0.4)	0	19 (1.6)	7 (-2.0)	-27 (-0.3)	33 (0.6)
GBP	-25 (-0.4)	-19 (-1.6)	0	-15 (-2.3)	-49 (-1.2)	15 (-1.2)
JPY	-10 (1.3)	-7 (2.0)	15 (2.3)	0	-32 (1.1)	26 (2.2)
AUD	24 (0.4)	27 (0.3)	49 (1.2)	32 (-1.1)	0	66 (0.6)
CAD	-37 (0.2)	-33 (-0.6)	-15 (1.2)	-26 (-2.2)	-66 (-0.6)	0

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research. Note: value in parenthesis is the 6m z-score, the color scheme is based on the z-score.

Figure 51. FX hedged yield pickup in the 10y sector using 3m forward hedging (annualized)

swapped ccy/asset ccy	*USTs	*Bunds	*Gilts	*JGBs	*AGBs	*CGBs
USD	0	5 (-0.3)	48 (1.4)	95 (-1.2)	-18 (0.3)	66 (0.5)
EUR	-5 (0.3)	0	43 (1.2)	94 (-0.5)	-20 (1.0)	64 (0.7)
GBP	-48 (-1.4)	-43 (-1.2)	0	49 (-1.7)	-65 (-0.5)	22 (-1.2)
JPY	-95 (1.2)	-94 (0.5)	-49 (1.7)	0	-111 (0.9)	-30 (1.3)
AUD	18 (-0.3)	20 (-1.0)	65 (0.5)	111 (-0.9)	0	89 (0.0)
CAD	-66 (-0.5)	-64 (-0.7)	-22 (1.2)	30 (-1.3)	-89 (0.0)	0

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research. Note: value in parenthesis is the 6m z-score, the color scheme is based on the z-score.

Figure 50. FX hedged yield pickup in the 5y sector using 3m forward hedging (annualized)

swapped ccy/asset ccy	*USTs	*Bunds	*Gilts	*JGBs	*AGBs	*CGBs
USD	0	3 (-0.5)	21 (0.1)	41 (-1.5)	-33 (-0.3)	52 (-0.3)
EUR	-3 (0.5)	0	19 (0.9)	42 (-1.9)	-33 (0.2)	52 (0.6)
GBP	-21 (-0.1)	-19 (-0.9)	0	21 (-2.2)	-54 (-0.5)	34 (-0.7)
JPY	-41 (1.5)	-42 (1.9)	-21 (2.2)	0	-73 (1.3)	10 (2.0)
AUD	33 (0.3)	33 (-0.2)	54 (0.5)	73 (-1.3)	0	90 (0.3)
CAD	-52 (0.3)	-52 (-0.6)	-34 (0.7)	-10 (-2.0)	-90 (-0.3)	0

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research. Note: value in parenthesis is the 6m z-score, the color scheme is based on the z-score.

Figure 52. FX hedged yield pickup in the 30y sector using 3m forward hedging (annualized)

swapped ccy/asset ccy	*USTs	*Bunds	*Gilts	*JGBs	*AGBs	*CGBs
USD	0	6 (-0.2)	64 (1.7)	168 (-0.5)	-12 (1.5)	65 (2.1)
EUR	-6 (0.2)	0	58 (0.9)	167 (-0.3)	-15 (1.4)	62 (1.3)
GBP	-64 (-1.7)	-58 (-0.9)	0	105 (-1.1)	-76 (0.5)	4 (0.8)
JPY	-168 (0.5)	-167 (0.3)	-105 (1.1)	0	-179 (1.2)	-105 (1.4)
AUD	12 (-1.5)	15 (-1.4)	76 (-0.5)	179 (-1.2)	0	82 (0.1)
CAD	-65 (-2.1)	-62 (-1.3)	-4 (-0.8)	105 (-1.4)	-82 (-0.1)	0

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research. Note: value in parenthesis is the 6m z-score, the color scheme is based on the z-score.

If you are visually impaired and would like to speak to a Citi representative regarding the details of the graphics in this document, please call USA 1-888-500-5008 (TTY: 711), from outside the US +1-210-677-3788

Appendix A-1

ANALYST CERTIFICATION

The research analysts primarily responsible for the preparation and content of this research report are either (i) designated by "AC" in the author block or (ii) listed in bold alongside content which is attributable to that analyst. If multiple AC analysts are designated in the author block, each analyst is certifying with respect to the entire research report other than (a) content attributable to another AC certifying analyst listed in bold alongside the content and (b) views expressed solely with respect to a specific issuer which are attributable to another AC certifying analyst identified in the price charts or rating history tables for that issuer shown below. Each of these analysts certify, with respect to the sections of the report for which they are responsible: (1) that the views expressed therein accurately reflect their personal views about each issuer and security referenced and were prepared in an independent manner, including with respect to Citigroup Global Markets Inc. and its affiliates; and (2) no part of the research analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in this report.

IMPORTANT DISCLOSURES

Citigroup Global Markets Inc. or its affiliates received compensation for products and services other than investment banking services from United States in the past 12 months.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, securities-related: United States.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, non-securities-related: United States.

Citigroup Global Markets Inc. and/or its affiliates has a significant financial interest in relation to United States. (For an explanation of the determination of significant financial interest, please refer to the policy for managing conflicts of interest which can be found at www.citiVelocity.com.)

Analysts' compensation is determined by Citi Research management and Citigroup's senior management and is based upon activities and services intended to benefit the investor clients of Citigroup Global Markets Inc. and its affiliates (the "Firm"). Compensation is not linked to specific transactions or recommendations. Like all Firm employees, analysts receive compensation that is impacted by overall Firm profitability which includes investment banking, sales and trading, and principal trading revenues. One factor in equity research analyst compensation is arranging corporate access events between institutional clients and the management teams of covered companies. Typically, company management is more likely to participate when the analyst has a positive view of the company.

For financial instruments recommended in the Product in which the Firm is not a market maker, the Firm is a liquidity provider in such financial instruments (and any underlying instruments) and may act as principal in connection with transactions in such instruments. The Firm is a regular issuer of traded financial instruments linked to securities that may have been recommended in the Product. The Firm regularly trades in the securities of the issuer(s) discussed in the Product. The Firm may engage in securities transactions in a manner inconsistent with the Product and, with respect to securities covered by the Product, will buy or sell from customers on a principal basis.

Unless stated otherwise neither the Research Analyst nor any member of their team has viewed the material operations of the Companies for which an investment view has been provided within the past 12 months.

For important disclosures (including copies of historical disclosures) regarding the companies that are the subject of this Citi Research product ("the Product"), please contact Citi Research, 388 Greenwich Street, 6th Floor, New York, NY, 10013, Attention: Legal/Compliance [E6WYB6412478]. In addition, the same important disclosures, with the exception of the Valuation and Risk assessments and historical disclosures, are contained on the Firm's disclosure website at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. Valuation and Risk assessments can be found in the text of the most recent research note/report regarding the subject company. Pursuant to the Market Abuse Regulation a history of all Citi Research recommendations published during the preceding 12-month period can be accessed via Citi Velocity (<https://www.citivelocity.com/cv2>) or your standard distribution portal. Historical disclosures (for up to the past three years) will be provided upon request.

RESEARCH ANALYST AFFILIATIONS / NON-US RESEARCH ANALYST DISCLOSURES

The legal entities employing the authors of this report are listed below (and their regulators are listed further herein). Non-US research analysts who have prepared this report (i.e., all research analysts listed below other than those identified as employed by Citigroup Global Markets Inc.) are not registered/qualified as research analysts with FINRA. Such research analysts may not be associated persons of the member organization (but are employed by an affiliate of the member organization) and therefore may not be subject to the FINRA Rule 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Citigroup Global Markets Inc.

Alejandra Vazquez Plata; Mike Chang; Jason Williams; Raghav Datla

OTHER DISCLOSURES

Any price(s) of instruments mentioned in recommendations are as of the prior day's market close on the primary market for the instrument, unless otherwise stated.

The completion and first dissemination of any recommendations made within this research report are as of the Eastern date-time displayed at the top of the Product. If the Product references views of other analysts then please refer to the price chart or rating history table for the date/time of completion and first dissemination with respect to that view.

European regulations require that where a recommendation differs from any of the author's previous recommendations concerning the same financial instrument or issuer that has been published during the preceding 12-month period that the change(s) and the date of that previous recommendation are indicated. Please refer to the trade history in the published research or contact the research analyst.

Citi Research has implemented policies for identifying, considering and managing potential conflicts of interest arising as a result of publication or distribution of investment research. A description of these policies can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

The proportion of all Citi Research research recommendations that were the equivalent to "Buy", "Hold", "Sell" at the end of each quarter over the prior 12 months (with the % of these that had received investment firm services from Citi in the prior 12 months shown in brackets) is as follows; Q2 2026 Buy 33%(63%), Hold 44% (51%), Sell 23% (49%), RV 0.5% (87%); Q1 2026 Buy 33% (63%), Hold 44% (52%), Sell 23% (46%), RV 0.5% (89%); Q4 2025 Buy 33% (63%), Hold 44% (50%), Sell 23% (46%), RV 0.4% (91%); Q3 2025 Buy 33%(61%), Hold 44% (52%), Sell 23% (50%), RV 0.4% (80%). For the purposes of disclosing recommendations other than for equity (whose definitions can be found in the corresponding disclosure sections), "Buy" means a positive directional trade idea; "Sell" means a negative directional trade idea; and "Relative Value" means any trade idea which does not have a clear direction to the investment strategy.

European regulations require a 5 year price history when past performance of a security is referenced. CitiVelocity's Charting Tool (https://www.citivelocity.com/cv2/#go/CHARTING_3_Equities) provides the facility to create customisable price charts including a five year option. This tool can be found in the Data & Analytics section under any of the asset class menus in CitiVelocity (<https://www.citivelocity.com/>). For further information contact CitiVelocity support (https://www.citivelocity.com/cv2/go/CLIENT_SUPPORT). The source for all referenced prices, unless otherwise stated, is DataCentral, which sources price information from LSEG Data & Analytics. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance.

Investors should always consider the investment objectives, risks, and charges and expenses of an ETF carefully before investing. The applicable prospectus and key investor information document (as applicable) for an ETF should contain this and other information about such ETF. It is important to read carefully any such prospectus before investing. Clients may obtain prospectuses and key investor information documents for ETFs from the applicable distributor or authorized participant, the exchange upon which an ETF is listed and/or from the applicable website of the applicable ETF issuer. The value of the investments and any accruing income may fall or rise. Any past performance, prediction or forecast is not indicative of future or likely performance. Any information on ETFs contained herein is provided strictly for illustrative purposes and should not be deemed an offer to sell or a solicitation of an offer to purchase units of any ETF either explicitly or implicitly. The opinions expressed are those of the authors and do not necessarily reflect the views of ETF issuers, any of their agents or their affiliates. Citigroup Global Markets India Private Limited and/or its affiliates may have, from time to time, actual or beneficial ownership of 1% or more in the debt securities of the subject issuer.

Please be advised that pursuant to Executive Order 13959 as amended (the "Order"), U.S. persons are prohibited from investing in securities of any company determined by the United States Government to be the subject of the Order. This research is not intended to be used or relied upon in any way that could result in a violation of the Order. Investors are encouraged to rely upon their own legal counsel for advice on compliance with the Order and other economic sanctions programs administered and enforced by the Office of Foreign Assets Control of the U.S. Treasury Department.

This communication is directed at persons who are "Eligible Clients" as such term is defined in the Israeli Regulation of Investment Advice, Investment Marketing and Investment Portfolio Management law, 1995 (the "Advisory Law"). Within Israel, this communication is not intended for retail clients and Citi will not make such products or transactions available to retail clients or to non-Eligible Clients. The presenter is not licensed as investment advisor or investment marketer by the Israeli Securities Authority ("ISA") and this communication does not constitute investment or marketing advice. The information contained herein may relate to matters that are not regulated by the ISA. Any securities which are the subject of this communication may not be offered or sold to any Israeli person except pursuant to a security offering exemption according to the Israeli Securities Law, 1968 and the public offering rules provided thereunder.

Citi Research broadly and simultaneously disseminates its research content to the Firm's institutional and retail clients via the Firm's proprietary electronic distribution platforms (e.g., Citi Velocity and various Global Wealth platforms). As a convenience, certain, but not all, research content may be distributed through third party aggregators. Clients may receive published research reports by email, on a discretionary basis, and only after such research content has been broadly disseminated. Certain research is made available only to institutional investors to satisfy regulatory requirements. The level and types of services provided by Citi Research analysts to clients may vary depending on various factors such as the client's individual preferences as to the frequency and manner of receiving communications from analysts, the client's risk profile and investment

focus and perspective (e.g. market-wide, sector specific, long term, short-term etc.), the size and scope of the overall client relationship with the Firm and legal and regulatory constraints.

Pursuant to Comissão de Valores Mobiliários Resolução 20 and ASIC Regulatory Guide 264, Citi is required to disclose whether a Citi related company or business has a commercial relationship with the subject company. Considering that Citi operates multiple businesses in more than 100 countries around the world, it is likely that Citi has a commercial relationship with the subject company.

Securities recommended, offered, or sold by the Firm: (i) are not insured by the Federal Deposit Insurance Corporation; (ii) are not deposits or other obligations of any insured depository institution (including Citibank); and (iii) are subject to investment risks, including the possible loss of the principal amount invested. The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security. Any decision to purchase securities mentioned in the Product must take into account existing public information on such security or any registered prospectus. Although information has been obtained from and is based upon sources that the Firm believes to be reliable, we do not guarantee its accuracy and it may be incomplete and condensed. Note, however, that the Firm has taken all reasonable steps to determine the accuracy and completeness of the disclosures made in the Important Disclosures section of the Product. The Firm's research department has received assistance from the subject company(ies) referred to in this Product including, but not limited to, discussions with management of the subject company(ies). Firm policy prohibits research analysts from sending draft research to subject companies. However, it should be presumed that the author of the Product has had discussions with the subject company to ensure factual accuracy prior to publication. Statements and views concerning ESG (environmental, social, governance) factors are typically based upon public statements made by the affected company or other public news, which the author may not have independently verified. ESG factors are one consideration that investors may choose to examine when making investment decisions. All opinions, projections and estimates constitute the judgment of the author as of the date of the Product and these, plus any other information contained in the Product, are subject to change without notice. Prices and availability of financial instruments also are subject to change without notice. Notwithstanding other departments within the Firm advising the companies discussed in this Product, information obtained in such role is not used in the preparation of the Product. Although Citi Research does not set a predetermined frequency for publication, if the Product is a fundamental equity or credit research report, it is the intention of Citi Research to provide research coverage of the covered issuers, including in response to news affecting the issuer. For non-fundamental research reports, Citi Research may not provide regular updates to the views, recommendations and facts included in the reports. Notwithstanding that Citi Research maintains coverage on, makes recommendations concerning or discusses issuers, Citi Research may be periodically restricted from referencing certain issuers due to legal or policy reasons. Where a component of a published trade idea is subject to a restriction, the trade idea will be removed from any list of open trade ideas included in the Product. Upon the lifting of the restriction, the trade idea will either be re-instated in the open trade ideas list if the analyst continues to support it or it will be officially closed. Citi Research may provide different research products and services to different classes of customers (for example, based upon long-term or short-term investment horizons) that may lead to differing conclusions or recommendations that could impact the price of a security contrary to the recommendations in the alternative research product, provided that each is consistent with the rating system for each respective product.

Investing in non-U.S. securities, including ADRs, may entail certain risks. The securities of non-U.S. issuers may not be registered with, nor be subject to the reporting requirements of the U.S. Securities and Exchange Commission. There may be limited information available on foreign securities. Foreign companies are generally not subject to uniform audit and reporting standards, practices and requirements comparable to those in the U.S. Securities of some foreign companies may be less liquid and their prices more volatile than securities of comparable U.S. companies. In addition, exchange rate movements may have an adverse effect on the value of an investment in a foreign stock and its corresponding dividend payment for U.S. investors. Net dividends to ADR investors are estimated, using withholding tax rates conventions, deemed accurate, but investors are urged to consult their tax advisor for exact dividend computations. Investors who have received the Product from the Firm may be prohibited in certain states or other jurisdictions from purchasing securities mentioned in the Product from the Firm. Please ask your Financial Consultant for additional details. Citigroup Global Markets Inc. takes responsibility for the Product in the United States. Any orders by US investors resulting from the information contained in the Product may be placed only through Citigroup Global Markets Inc.

The Citigroup legal entity that takes responsibility for the production of the Product is the legal entity which the first named author is employed by.

The Product is made available in **Australia** through Citigroup Global Markets Australia Pty Limited. (ABN 64 003 114 832 and AFSL No. 240992), participant of the ASX Group and regulated by the Australian Securities & Investments Commission. Citigroup Centre, 2 Park Street, Sydney, NSW 2000. Citigroup Global Markets Australia Pty Limited is not an Authorised Deposit-Taking Institution under the Banking Act 1959, nor is it regulated by the Australian Prudential Regulation Authority.

The Product is made available in **Brazil** by Citigroup Global Markets Brasil - CCTVM SA, which is regulated by CVM - Comissão de Valores Mobiliários ("CVM"), BACEN - Brazilian Central Bank, APIMEC - Associação dos Analistas e Profissionais de Investimento do Mercado de Capitais and ANBIMA - Associação Brasileira das Entidades dos Mercados Financeiro e de Capitais. Av. Paulista, 1111 - 14º andar(parte) - CEP: 01311920 - São Paulo - SP.

This Product is available in **Chile** through Banchile Corredores de Bolsa S.A., an indirect subsidiary of Citigroup Inc., which is regulated by the Comisión Para El Mercado Financiero. Enrique Foster Sur, 20, piso 6, Las Condes, Santiago, Chile.

Disclosure for investors in the Republic of Colombia : This communication or message does not constitute a professional recommendation to make investment in the terms of article 2.40.1.1.2 of Decree 2555 de 2010 or the regulations that modify,

substitute or complement it. Para la elaboración y distribución de informes de investigación y de comunicaciones generales de que trata este artículo no se requiere ser una entidad vigilada por la Financiera de Colombia.

The Product is made available in **Germany** by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt für
tungaufsicht BaFin). Börsenplatz 9, 60313 Frankfurt am Main, Germany.

Unless otherwise specified, if the analyst who prepared this report is based in Hong Kong and it relates to "securities" (as defined in the Securities and Futures Ordinance (Cap.571 of the Laws of Hong Kong)), the report is issued in Hong Kong by Citigroup Global Markets Asia Limited. Citigroup Global Markets Asia Limited is regulated by Hong Kong Securities and Futures Commission. If the report is prepared by a non-Hong Kong based analyst, please note that such analyst (and the legal entity that the analyst is employed by or accredited to) is not licensed/registered in Hong Kong and they do not hold themselves out as such. Please refer to the section "Research Analyst Affiliations / Non-US Research Analyst Disclosures" for the details of the employment entity of the analysts.

The Product is made available in **India** by Citigroup Global Markets India Private Limited (CGM), which is regulated by the Securities and Exchange Board of India (SEBI), as a Research Analyst (SEBI Registration No. INH000000438). CGM is also actively involved in the business of merchant banking (SEBI Registration No. INM000010718) and stock brokerage ((SEBI Registration No. INZ000263033) in India, and is registered with SEBI in this regard. Registration granted by SEBI, enlistment with Bombay Stock Exchange (BSE) and certification from National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. CGM's registered office is at 1202, 12th Floor, First International Financial Centre (IFIC), G Block, Bandra Kurla Complex, Bandra East, Mumbai – 400098 & registered Tel: +91 22 61759999. Citi maintains robust policies, procedures, controls, and training to ensure continued compliance with all applicable rules and regulations. All recommendations contained herein are made by duly qualified research analysts. CGM's Corporate Identity Number is 26657, and its Compliance Officer [Vishal Bohra] contact details are: Tel: +91-022-61759994, Fax: +91-022-61759851, Email: cgmcompliance@citi.com. The Investor Charter in respect of Research Analysts, the Compliance Audit Report and Complaints information can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. The grievance officer [Niraj Mody] contact details are Tel: +91-22-42775002, Email: EMEA.CR.Complaints@citi.com. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. SEBI prescribed Client Terms & Conditions can be found at <https://www.citivelocity.com/rendition/VXNILmh0bWw/eppublic/V1/file?paramData=L0dETVMvcHVj>

The Product is made available in **Indonesia** through PT Citigroup Sekuritas Indonesia. Citibank Tower 10/F, Pacific Century Place, SCBD lot 10, Jl. Jend Sudirman Kav 52-53, Jakarta 12190, Indonesia. Neither this Product nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens wherever they are domiciled or to Indonesian residents except in compliance with applicable capital market laws and regulations. This Product is not an offer of securities in Indonesia. The securities referred to in this Product have not been registered with the Capital Market and Financial Services Authority (OJK) pursuant to relevant capital market laws and regulations, and may not be offered or sold within the territory of the Republic of Indonesia or to Indonesian citizens through a public offering or in circumstances which constitute an offer within the meaning of the Indonesian capital market laws and regulations.

The Product is made available in **Japan** by Citigroup Global Markets Japan Inc. ("CGMJ"), which is regulated by Financial Services Agency, Securities and Exchange Surveillance Commission, Japan Securities Dealers Association, Tokyo Stock Exchange and Osaka Securities Exchange. Otemachi Park Building, 1-1-1 Otemachi, Chiyoda-ku, Tokyo 100-8132 Japan. In the event that an error is found in an CGMJ research report, a revised version will be posted on the Firm's Citi Velocity website. If you have questions regarding Citi Velocity, please call (81 3) 6270-3019 for help.

The product is made available in the **Kingdom of Saudi Arabia** in accordance with Saudi laws through Citigroup Saudi Arabia, which is regulated by the Capital Market Authority (CMA) under CMA license (17184-31). 2239 Al Urubah Rd – Al Olaya Dist. Unit No. 18, Riyadh 12214 – 9597, Kingdom Of Saudi Arabia.

The Product is made available in **Korea** by Citigroup Global Markets Korea Securities Ltd. (CGMK), which is regulated by the Financial Services Commission, the Financial Supervisory Service and the Korea Financial Investment Association (KOFIA). The address of CGMK is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. KOFIA makes available registration information of research analysts on its website. Please visit the following website if you wish to find KOFIA registration information on research analysts of

CGMK. <http://dis.kofia.or.kr/websquare/index.jsp?w2xPath=/wq/fundMgr/tList.xml&divisionId=MDISO300200200000&serviceld=SDISO3002002000>. The Product is made available in Korea by Citibank Korea Inc., which is regulated by the Financial Services Commission and the Financial Supervisory Service. Address is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. This research report is intended to be provided only to Professional Investors as defined in the Financial Investment Services and Capital Market Act and its Enforcement Decree in Korea.

The Product is made available in **Malaysia** by Citigroup Global Markets Malaysia Sdn Bhd (Registration No. 199801004692 (460819-D)) ("CGMM") to its clients and CGMM takes responsibility for its contents as regards CGMM's clients. CGMM is regulated by the Securities Commission Malaysia. Please contact CGMM at Level 43 Menara Citibank, 165 Jalan Ampang, 50450 Kuala Lumpur, Malaysia in respect of any matters arising from, or in connection with, the Product.

The Product is made available in **Mexico** by Citi México Casa de Bolsa, S.A. de C.V., Grupo Financiero Citi México which is a wholly owned subsidiary of Citigroup Inc. and is regulated by Comisión Nacional Bancaria y de Valores. Prolongación Reforma 1196, 24 floor, Colonia Santa Fe, Alcaldía Cuajimalpa de Morelos, C.P. 05348, Ciudad de México.

The Product is made available in **Poland** by Biuro Maklerskie Banku Handlowego (DMBH), separate department of Bank Handlowy w Warszawie S.A. a subsidiary of Citigroup Inc., which is regulated by Komisja Nadzoru Finansowego. Biuro Maklerskie Banku Handlowego (DMBH), ul.Senatorska 16, 00-923 Warszawa.

The Product is made available in **Singapore** through Citigroup Global Markets Singapore Pte. Ltd. ("CGMSPL"), a capital markets services and Exempt Financial Advisor license holder, and regulated by Monetary Authority of Singapore. Please contact CGMSPL at 8 Marina View, 21st Floor Asia Square Tower 1, Singapore 018960, in respect of any matters arising from, or in connection with, the analysis of this document. This Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citi Private Bank, the Product is made available in Singapore by Citi Private Bank through Citibank, N.A., Singapore Branch. Citibank N.A., Singapore Branch is a licensed bank in Singapore that is regulated by the Monetary Authority of Singapore. Please contact your Private Banker in Citibank N.A., Singapore Branch if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citibank Singapore Limited ("CSL"), the Product is distributed in Singapore by CSL to selected Citigold/Citigold Private Clients. CSL provides no independent research or analysis of the substance or in preparation of the Product. Please contact your Citigold/Citigold Private Client Relationship Manager in CSL if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited investors as defined under the Securities and Futures Act (Cap. 289).

Citigroup Global Markets (Pty) Ltd. is incorporated in the **Republic of South Africa** (company registration number 2000/025866/07) and its registered office is at 145 West Street, Sandton, 2196, Saxonwold. Citigroup Global Markets (Pty) Ltd. is regulated by JSE Securities Exchange South Africa, South African Reserve Bank and the Financial Services Board. The investments and services contained herein are not available to private customers in South Africa.

The Product is made available in the **Republic of China (Taiwan)** through Citigroup Global Markets Taiwan Securities Company Ltd. ("CGMTS"), 14F, 15F and 16F, No. 1, Songzhi Road, Taipei 110, Taiwan, subject to the license scope and the applicable laws and regulations in the Republic of China (Taiwan). CGMTS is regulated by the Securities and Futures Bureau of the Financial Supervisory Commission of Taiwan, the Republic of China (Taiwan). No portion of the Product may be reproduced or quoted in the Republic of China (Taiwan) by the press or any third parties [without the written authorization of CGMTS]. Pursuant to the applicable laws and regulations in the Republic of China (Taiwan), the recipient of the Product shall not take advantage of such Product to involve in any matters in which the recipient may have conflicts of interest. If the Product covers securities which are not allowed to be offered or traded in the Republic of China (Taiwan), neither the Product nor any information contained in the Product shall be considered as advertising the securities or making recommendation of the securities in the Republic of China (Taiwan). The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security or financial products. Any decision to purchase securities or financial products mentioned in the Product must take into account existing public information on such security or the financial products or any registered prospectus.

The Product is made available in **Thailand** through Citicorp Securities (Thailand) Ltd., which is regulated by the Securities and Exchange Commission of Thailand. 399 Interchange 21 Building, 18th Floor, Sukhumvit Road, Klongtoey Nua, Wattana ,Bangkok 10110, Thailand.

Disclosure for investors in the Republic of Turkey: The Product is made available in **Turkey** through Citibank AS which is regulated by Capital Markets Board. Tekfen Tower, Eski Buyukdere Caddesi # 209 Kat 2B, 23294 Levent, Istanbul, Turkey. Under Capital Markets Law of Turkey (Law No: 6362), the investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Comments and recommendations stated here rely on the individual opinions of the ones providing these comments and recommendations. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations. Furthermore, Citi Research is a division of Citigroup Global Markets Inc. (the "Firm"), which does and seeks to do business with companies and/or trades on securities covered in this research reports. As a result, investors should be aware that the Firm may have a conflict of interest that could affect the objectivity of this report, however investors should also note that the Firm has in place organisational and administrative arrangements to manage potential conflicts of interest of this nature.

In the **U.A.E**, these materials (the "Materials") are communicated by Citigroup Global Markets Limited, DIFC branch ("CGML"), an in the Dubai International Financial Center ("DIFC") and licensed and regulated by the Dubai Financial Services Authority ("DFSA", license #CLO221) to Professional Clients and Market Counterparties, as defined in DFSA regulations, only and should not be relied upon or distributed to Retail Clients. Financial products and/or services to which the Materials relate will only be made available to Professional Clients and ties. Citigroup Global Markets Limited DIFC Branch registered address is Level 3, Gate District Building 02, Dubai International Financial Centre and can be contacted on +971 4 509 97 90.

The Product is made available in **United Kingdom** by Citigroup Global Markets Limited, which is authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA. This material may relate to investments or services of a person outside of the UK or to other matters which are not authorised by the PRA nor regulated by the FCA and the PRA and further details as to where this may be the case are available upon request in respect of this material. Citigroup Centre, Canada Square, Canary Wharf, London, E14 5LB.

The Product is made available in **United States** and **Canada** by Citigroup Global Markets Inc., which is a member of FINRA and registered with the US Securities and Exchange Commission. 388 Greenwich Street, New York, NY 10013.

Unless specified to the contrary, within EU Member States, the Product is made available by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt für tungsaufsicht-BaFin).

The Product is not to be construed as providing investment services in any jurisdiction where the provision of such services would not be permitted.

Subject to the nature and contents of the Product, the investments described therein are subject to fluctuations in price and/or value and investors may get back less than originally invested. Certain high-volatility investments can be subject to sudden and large falls in value that could equal or exceed the amount invested. The yield and average life of CMOs (collateralized mortgage obligations) referenced in this Product will fluctuate depending on the actual rate at which mortgage holders prepay the mortgages underlying the CMO and changes in current interest rates. Any government agency backing of the CMO applies only to the face value of the CMO and not to any premium paid. Certain investments contained in the Product may have tax implications for private customers whereby levels and basis of taxation may be subject to change. If in doubt, investors should seek advice from a tax adviser. The Product does not purport to identify the nature of the specific market or other risks associated with a particular transaction. Advice in the Product is general and should not be construed as personal advice given it has been prepared without taking account of the objectives, financial situation or needs of any particular investor. Accordingly, investors should, before acting on the advice, consider the appropriateness of the advice, having regard to their objectives, financial situation and needs. Prior to acquiring any financial product, it is the client's responsibility to obtain the relevant offer document for the product and consider it before making a decision as to whether to purchase the product.

Card Insights. Where this report references Card Insights data, Card Insights consists of selected data from a subset of Citi's proprietary credit card transactions. Such data has undergone rigorous security protocols to keep all customer information confidential and secure; the data is highly aggregated and anonymized so that all unique customer identifiable information is removed from the data prior to receipt by the report's author or distribution to external parties. This data should be considered in the context of other economic indicators and publicly available information. Further, the selected data represents only a subset of Citi's proprietary credit card transactions due to the selection methodology or other limitations and should not be considered as indicative or predictive of the past or future financial performance of Citi or its credit card business.

Citi Research product may source data from dataCentral. dataCentral is a Citi Research proprietary database, which includes the Firm's estimates, data from company reports and feeds from LSEG Data & Analytics. The source for all referenced prices, unless otherwise stated, is DataCentral. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance. The printed and printable version of the research report may not include all the information <(e.g. certain financial summary information and comparable company data) that is linked to the online version available on the Firm's proprietary electronic distribution platforms.

Where included in this report, MSCI sourced information is the exclusive property of Morgan Stanley Capital International Inc. (MSCI). Without prior written permission of MSCI, this information and any other MSCI intellectual property may not be reproduced, disseminated or used to create any financial products, including any indices. This information is provided on an "as is" basis. The user assumes the entire risk of any use made of this information. MSCI, its affiliates and any third party involved in, or related to, computing or compiling the information hereby expressly disclaim all warranties of originality, accuracy, completeness, merchantability or fitness for a particular purpose with respect to any of this information. Without limiting any of the foregoing, in no event shall MSCI, any of its affiliates or any third party involved in, or related to, computing or compiling the information have any liability for any damages of any kind. MSCI, Morgan Stanley Capital International and the MSCI indexes are services marks of MSCI and its affiliates. Where data is attributed to Morningstar that data is © 2026 Morningstar, Inc. All Rights Reserved. That information: (1) is proprietary to Morningstar and/or its content providers; (2) may not be copied or distributed; and (3) is not warranted to be accurate, complete or timely. Neither Morningstar nor its content providers are responsible for any damages or losses arising from any use of this information.

The Firm accepts no liability whatsoever for the actions of third parties. The Product may provide the addresses of, or contain hyperlinks to, websites. Except to the extent to which the Product refers to website material of the Firm, the Firm has not reviewed the linked site. Equally, except to the extent to which the Product refers to website material of the Firm, the Firm takes no responsibility for, and makes no representations or warranties whatsoever as to, the data and information contained therein. Such address or hyperlink (including addresses or hyperlinks to website material of the Firm) is provided solely for your convenience and information and the content of the linked site does not in any way form part of this document. Accessing such website or following such link through the Product or the website of the Firm shall be at your own risk and the Firm shall have no liability arising out of, or in connection with, any such referenced website.

© 2026 Citigroup Global Markets Inc. Citi Research is a division of Citigroup Global Markets Inc. Citi and Citi and Arc Design are trademarks and service marks of Citigroup Inc. and its affiliates and are used and registered throughout the world. All rights reserved. The research data in this report are not intended to be used for the purpose of (a) determining the price of or amounts due in respect of (or to value) one or more financial products or instruments and/or (b) measuring or comparing the performance of, or defining the asset allocation of a financial product, a portfolio of financial instruments, or a collective investment undertaking, and any such use is strictly prohibited without the prior written consent of Citi Research. Any unauthorized use, duplication, redistribution or disclosure of this report (the "Product"), including, but not limited to, redistribution of the Product by electronic mail, posting of the Product on a website or page, and/or providing to a third party a link to the Product, is prohibited by law and will result in prosecution. The information contained in the Product is intended solely for the recipient and may not be further distributed by the recipient to any third party.

ADDITIONAL INFORMATION IS AVAILABLE UPON REQUEST
